Supplementary Material — Group-conformal calibration of near-infrared prediction intervals for new seasons, instruments and regions using PLS diagnostics

[Authors]

Abstract

Supplementary tables S1–S28 and figures S1–S3. All values are generated from the result files of the accompanying code (run_all.py).

Keywords: Supplementary material

S1. Method settings

Table S1. Fixed settings of all experiments (config/experiment_config.json).

Component	Setting
seeds	[0, 1, 2, 3, 4]
alphas	[0.05, 0.1, 0.2]
primary_alpha	0.1
preprocessing	{mango: Savitzky-Golay 2nd derivative, window 13, poly 2, wavelengths 684-990 nm, ossl: log10(1/R) at 10 nm steps 400-2500 nm, Savitzky-Golay 1st derivative, window 11; target log(1+OC), corn: SNV then Savitzky-Golay 1st derivative, window 15, tablets: SNV then Savitzky-Golay 1st derivative, window 15}
pls	{A_max: {mango: 20, ossl: 25, corn: 15, tablets: 15}, selection: min group-balanced RMSE of out-of-group cross-validation}
split_conformal	{calibration_fraction: 0.25, split_unit: physical unit}
random_cv_folds	5
scale_model	{features: [log(1+T2), log(Q/Q_ref)], link: log, loss: Poisson quasi-likelihood, floor_frac: 0.1}
gpr	{subset: 1000, kernel: Constant*RBF(ARD)+White}
bagging	{B: 25, bootstrap_unit: physical unit}
quantile_gbm	{n_estimators: 300, learning_rate: 0.05, num_leaves: 31}
wcp	{density_ratio: logistic regression on standardised PLS scores and log Q, max_n_per_class: 20000}
lucas_spatial_blocks	{k: 30, method: k-means on (lat, lon*cos(lat)), seed: 0}
cnn	{architecture: Conv1d(1->8, kernel 9, same padding) - ELU - Flatten - Linear(64) - ELU - Linear(32) - ELU - Linear(1), inputs: per-wavelength standardised (training statistics), optimizer: Adam lr 1e-3, weight decay 1e-4, batch 256, epochs: max 40, early stopping patience 8 on a 10 % unit-grouped internal validation split, device: auto (CPU/CUDA/MPS)}
wcp_clip	{weights: density ratios normalised to mean 1 on the calibration set and capped at 20}

S2. Generalisation scenarios and exchangeable control

Table S2. Coverage (cov.) and interval score (IS) at nominal 0.90 for the LUCAS campaign shift (2009 ↔ 2015), the KSSL → LUCAS library transfer, and the tablet data (calibration on instrument 1; test on instrument 1 = exchangeable control, instrument 2 = instrument shift without group information). Means over five seeds.

Method	S-campaign cov.	S-campaign IS	S-KSSL→LUCAS cov.	S-KSSL→LUCAS IS	Tablets inst. 1 cov.	Tablets inst. 2 cov.
ASTM-R	0.861	13.137	0.053	136.577	0.974	0.890
ASTM-G	0.876	12.788	0.071	126.234	0.974	0.890
BAG	0.861	13.154	0.055	136.924	0.976	0.910
GPR	0.802	12.546	0.975	83.180	0.962	0.749
CQR	0.874	9.760	0.310	105.562	0.920	0.961
SCP	0.847	13.591	0.046	148.382	0.932	0.760
CV+	0.852	13.352	0.050	139.221	0.948	0.831
WCP	0.803	∞	1.000	∞	0.923	1.000
G-W-A	0.869	12.945	0.066	128.916	0.940	0.795
GC-D	0.892	12.248	0.604	95.520	0.936	0.889
GC-D+	0.924	11.856	0.663	99.205	0.940	0.893
HJ+	0.908	12.169	0.078	122.659	0.948	0.831

Method	S-campaign cov.	S-campaign IS	S-KSSL→LUCAS cov.	S-KSSL→LUCAS IS	Tablets inst. 1 cov.	Tablets inst. 2 cov.
HCP-D	0.955	12.715	0.712	101.500	0.947	0.926
ORACLE	0.902	11.382	0.902	134.476	0.901	0.902

S3. Other nominal levels

Table S3a. Group-averaged coverage at nominal 0.95.

Method	M-season	M-forward	M-instr.	M-pop.	S-region	C-moist	C-oil	C-prot	C-starch	S-campaign	S-KSSL→LUCAS	Tablets
ASTM-R	0.893	0.860	0.871	0.939	0.932	0.649	0.883	0.773	0.661	0.911	0.089	0.950
ASTM-G	0.945	0.929	0.933	0.946	0.943	0.653	0.933	0.812	0.707	0.922	0.127	0.950
BAG	0.893	0.862	0.874	0.939	0.933	0.658	0.926	0.814	0.736	0.911	0.094	0.963
GPR	0.866	0.828	0.828	0.925	0.928	0.719	0.883	0.819	0.697	0.863	0.994	0.909
QGB	0.827	0.764	0.819	0.867	0.901	0.812	0.782	0.767	0.757	0.884	0.315	0.878
CQR	0.875	0.827	0.871	0.909	0.942	0.958	0.966	0.966	0.948	0.932	0.337	0.971
SCP	0.896	0.865	0.873	0.941	0.940	0.663	0.888	0.732	0.708	0.915	0.092	0.905
NCP-kNN	0.857	0.813	0.819	0.909	0.932	0.656	0.888	0.739	0.701	0.879	0.201	0.957
CV+	0.896	0.862	0.874	0.941	0.940	0.654	0.890	0.760	0.698	0.917	0.100	0.942
WCP	0.930	0.925	0.939	0.950	0.942	0.986	0.973	0.997	0.970	0.833	1.000	0.980
HJ+	1.000	1.000	0.977	0.955	0.982	1.000	1.000	1.000	1.000	0.972	0.214	0.942
HCP-D	1.000	1.000	1.000	0.961	1.000	1.000	1.000	1.000	1.000	1.000	0.948	0.984
GC-D+	1.000	1.000	0.972	0.954	0.982	1.000	1.000	1.000	1.000	0.978	0.926	0.963
GC-D	0.947	0.933	0.934	0.949	0.949	0.569	0.834	0.777	0.742	0.943	0.878	0.959
ORACLE	0.949	0.949	0.950	0.948	0.952	1.000	1.000	1.000	1.000	0.953	0.952	0.952

The difference GC-D − SCP for every scenario and nominal level is listed in results/revision/alpha_consistency.csv; it has the same sign at 0.80, 0.90 and 0.95 in every mango and soil scenario.

Table S3b. Group-averaged coverage at nominal 0.80.

Method	M-season	M-forward	M-instr.	M-pop.	S-region	C-moist	C-oil	C-prot	C-starch	S-campaign	S-KSSL→LUCAS	Tablets
ASTM-R	0.732	0.674	0.688	0.803	0.807	0.583	0.727	0.556	0.518	0.767	0.028	0.877
ASTM-G	0.821	0.790	0.789	0.819	0.827	0.587	0.783	0.621	0.568	0.788	0.036	0.877
BAG	0.731	0.678	0.690	0.803	0.809	0.583	0.761	0.586	0.547	0.765	0.029	0.896
GPR	0.685	0.634	0.640	0.779	0.800	0.610	0.709	0.600	0.548	0.699	0.916	0.768
QGB	0.607	0.526	0.572	0.676	0.721	0.610	0.625	0.610	0.565	0.691	0.205	0.734
CQR	0.659	0.587	0.627	0.730	0.779	0.784	0.781	0.824	0.736	0.767	0.222	0.877
SCP	0.714	0.661	0.667	0.785	0.778	0.558	0.724	0.576	0.522	0.725	0.021	0.711
NCP-kNN	0.686	0.640	0.645	0.750	0.771	0.553	0.725	0.579	0.531	0.679	0.062	0.802
CV+	0.713	0.658	0.669	0.785	0.779	0.573	0.739	0.564	0.526	0.735	0.024	0.785
WCP	0.785	0.786	0.839	0.825	0.866	0.927	0.896	0.967	0.870	0.761	1.000	0.922
HJ+	0.934	0.978	0.819	0.806	0.827	1.000	1.000	1.000	1.000	0.789	0.028	0.785
HCP-D	1.000	1.000	0.829	0.811	0.854	1.000	1.000	1.000	1.000	0.837	0.357	0.857
GC-D+	0.929	0.968	0.804	0.804	0.824	1.000	1.000	1.000	1.000	0.816	0.242	0.834

Method	M-season	M-forward	M-instr.	M-pop.	S-region	C-moist	C-oil	C-prot	C-starch	S-campaign	S-KSSL→LUCAS	Tablets
GC-D	0.797	0.770	0.776	0.799	0.798	0.458	0.641	0.621	0.602	0.787	0.219	0.821
ORACLE	0.799	0.798	0.800	0.800	0.802	0.826	0.815	0.812	0.813	0.802	0.801	0.801

The difference GC-D − SCP for every scenario and nominal level is listed in results/revision/alpha_consistency.csv; it has the same sign at 0.80, 0.90 and 0.95 in every mango and soil scenario.

S4. Worst-group coverage and share of under-covered groups

Table S4a. Lowest coverage among held-out groups (nominal 0.90).

Method	M-season	M-forward	M-instr.	M-pop.	S-region	C-moist	C-oil	C-prot	C-starch	S-campaign	S-KSSL→LUCAS	Tablets
ASTM-R	0.733	0.699	0.251	0.013	0.764	0.000	0.270	0.090	0.000	0.644	0.006	0.890
ASTM-G	0.798	0.811	0.336	0.013	0.794	0.000	0.530	0.120	0.000	0.659	0.010	0.890
BAG	0.733	0.703	0.258	0.013	0.768	0.000	0.540	0.150	0.000	0.645	0.006	0.910
GPR	0.659	0.653	0.157	0.010	0.758	0.090	0.550	0.150	0.000	0.442	0.879	0.749
QGB	0.616	0.565	0.303	0.013	0.752	0.560	0.560	0.480	0.340	0.647	0.067	0.759
CQR	0.675	0.644	0.346	0.037	0.816	0.820	0.700	0.770	0.490	0.739	0.075	0.920
SCP	0.721	0.695	0.236	0.013	0.759	0.000	0.220	0.060	0.000	0.627	0.004	0.760
NCP-kNN	0.674	0.581	0.137	0.345	0.791	0.000	0.230	0.090	0.000	0.585	0.020	0.865
CV+	0.724	0.692	0.244	0.013	0.757	0.000	0.400	0.080	0.000	0.632	0.006	0.831
WCP	0.717	0.682	0.567	0.028	0.599	0.830	0.800	0.950	0.700	0.560	1.000	0.923
HJ+	1.000	1.000	0.452	0.013	0.843	1.000	1.000	1.000	1.000	0.720	0.012	0.831
HCP-D	1.000	1.000	0.600	0.015	0.885	1.000	1.000	1.000	1.000	0.809	0.272	0.926
GC-D+	1.000	1.000	0.447	0.013	0.837	1.000	1.000	1.000	1.000	0.720	0.114	0.893
GC-D	0.798	0.772	0.356	0.013	0.786	0.000	0.330	0.140	0.020	0.649	0.092	0.889
ORACLE	0.896	0.896	0.897	0.868	0.897	0.880	0.840	0.840	0.870	0.896	0.898	0.901

Table S4b. Share of held-out groups with coverage below 0.85 (nominal 0.90).

Method	M-season	M-forward	M-instr.	M-pop.	S-region	C-moist	C-oil	C-prot	C-starch	S-campaign	S-KSSL→LUCAS	Tablets
ASTM-R	0.57	0.80	0.46	0.23	0.27	0.37	0.40	0.53	0.43	0.40	1.00	0.00
ASTM-G	0.14	0.40	0.17	0.18	0.23	0.33	0.23	0.43	0.33	0.34	1.00	0.00
BAG	0.57	0.80	0.46	0.22	0.27	0.33	0.30	0.50	0.33	0.40	1.00	0.00
GPR	0.71	0.80	0.62	0.26	0.30	0.37	0.43	0.63	0.70	0.59	0.00	0.50
QGB	1.00	1.00	0.92	0.55	0.60	0.90	0.90	0.90	0.97	0.78	1.00	0.50
CQR	0.57	1.00	0.67	0.35	0.17	0.07	0.20	0.13	0.47	0.24	1.00	0.00
SCP	0.71	0.80	0.54	0.24	0.27	0.37	0.60	0.57	0.63	0.45	1.00	0.50
NCP-kNN	0.71	1.00	0.75	0.37	0.27	0.40	0.47	0.50	0.60	0.76	1.00	0.00
CV+	0.71	0.80	0.58	0.24	0.27	0.37	0.43	0.53	0.47	0.41	1.00	0.50
WCP	0.29	0.20	0.21	0.16	0.10	0.03	0.07	0.00	0.13	0.64	0.00	0.00
HJ+	0.00	0.00	0.08	0.18	0.07	0.00	0.00	0.00	0.00	0.19	1.00	0.50
HCP-D	0.00	0.00	0.08	0.17	0.00	0.00	0.00	0.00	0.00	0.02	0.90	0.00
GC-D+	0.00	0.00	0.17	0.18	0.13	0.00	0.00	0.00	0.00	0.09	0.83	0.00

Method	M-season	M-forward	M-instr.	M-pop.	S-region	C-moist	C-oil	C-prot	C-starch	S-campaign	S-KSSL→LUCAS	Tablets
GC-D	0.29	0.40	0.29	0.19	0.27	0.63	0.57	0.67	0.53	0.29	0.90	0.00
ORACLE	0.00	0.00	0.00	0.00	0.00	0.00	0.03	0.03	0.00	0.00	0.00	0.00

S5. Sensitivity analysis

Table S5. Sensitivity of coverage, width and interval score to the number of latent variables (A_offset), the features of the scale model (diag_mode), the floor of σ (floor_frac), the preprocessing, and the number of population folds (nominal 0.90, seed 0).

task	variant	method	coverage	width	interval_score	A
mango_population	group_folds=20	G-W-A	0.900	3.694	4.959	19.201
mango_population	group_folds=20	GC-D	0.898	3.680	4.921	19.201
mango_population	group_folds=20	R-U-D	0.891	3.571	4.920	19.201
mango_population	group_folds=20	SCP	0.885	3.530	4.977	19.201
mango_population	group_folds=5	G-W-A	0.901	3.705	4.955	19.201
mango_population	group_folds=5	GC-D	0.900	3.692	4.920	19.201
mango_population	group_folds=5	R-U-D	0.891	3.569	4.906	19.201
mango_population	group_folds=5	SCP	0.885	3.532	4.969	19.201
mango_season_loso	A_offset=-4	G-W-A	0.900	4.832	6.297	10.857
mango_season_loso	A_offset=-4	GC-D	0.894	4.836	6.249	10.857
mango_season_loso	A_offset=-4	R-U-D	0.852	4.203	6.309	10.857
mango_season_loso	A_offset=-4	SCP	0.837	4.051	6.484	10.857
mango_season_loso	A_offset=4	G-W-A	0.894	4.532	6.313	18.286
mango_season_loso	A_offset=4	GC-D	0.884	4.498	6.081	18.286
mango_season_loso	A_offset=4	R-U-D	0.825	3.717	6.406	18.286
mango_season_loso	A_offset=4	SCP	0.806	3.515	6.719	18.286
mango_season_loso	diag_mode=Q	G-W-A	0.899	4.507	6.157	14.857
mango_season_loso	diag_mode=Q	GC-D	0.898	4.561	6.003	14.857
mango_season_loso	diag_mode=Q	R-U-D	0.840	3.789	6.194	14.857
mango_season_loso	diag_mode=Q	SCP	0.823	3.602	6.395	14.857
mango_season_loso	diag_mode=T2	G-W-A	0.899	4.507	6.157	14.857
mango_season_loso	diag_mode=T2	GC-D	0.894	4.512	5.939	14.857
mango_season_loso	diag_mode=T2	R-U-D	0.829	3.706	6.241	14.857
mango_season_loso	diag_mode=T2	SCP	0.823	3.602	6.395	14.857

task	variant	method	coverage	width	interval_score	A
mango_season_loso	floor_frac=0.02	G-W-A	0.899	4.507	6.157	14.857
mango_season_loso	floor_frac=0.02	GC-D	0.895	4.519	5.900	14.857
mango_season_loso	floor_frac=0.02	R-U-D	0.839	3.787	6.129	14.857
mango_season_loso	floor_frac=0.02	SCP	0.823	3.602	6.395	14.857
mango_season_loso	floor_frac=0.3	G-W-A	0.899	4.507	6.157	14.857
mango_season_loso	floor_frac=0.3	GC-D	0.895	4.519	5.900	14.857
mango_season_loso	floor_frac=0.3	R-U-D	0.839	3.787	6.129	14.857
mango_season_loso	floor_frac=0.3	SCP	0.823	3.602	6.395	14.857
mango_season_loso	preprocessing=SNV+SG1	G-W-A	0.898	4.893	6.549	19.286
mango_season_loso	preprocessing=SNV+SG1	GC-D	0.893	4.899	6.501	19.286
mango_season_loso	preprocessing=SNV+SG1	R-U-D	0.833	4.073	6.710	19.286
mango_season_loso	preprocessing=SNV+SG1	SCP	0.823	3.961	6.838	19.286
ossl_lucas_block	A_offset=-4	G-W-A	0.900	6.545	12.883	20.900
ossl_lucas_block	A_offset=-4	GC-D	0.898	7.272	12.246	20.900
ossl_lucas_block	A_offset=-4	R-U-D	0.891	7.076	12.307	20.900
ossl_lucas_block	A_offset=-4	SCP	0.886	6.163	13.088	20.900
ossl_lucas_block	A_offset=4	G-W-A	0.899	6.467	12.483	25.000
ossl_lucas_block	A_offset=4	GC-D	0.899	7.140	11.893	25.000
ossl_lucas_block	A_offset=4	R-U-D	0.889	6.849	11.988	25.000
ossl_lucas_block	A_offset=4	SCP	0.883	6.083	12.850	25.000
ossl_lucas_block	diag_mode=Q	G-W-A	0.899	6.470	12.496	24.900
ossl_lucas_block	diag_mode=Q	GC-D	0.898	6.951	11.979	24.900
ossl_lucas_block	diag_mode=Q	R-U-D	0.888	6.646	12.123	24.900
ossl_lucas_block	diag_mode=Q	SCP	0.883	6.086	12.855	24.900
ossl_lucas_block	diag_mode=T2	G-W-A	0.899	6.470	12.496	24.900
ossl_lucas_block	diag_mode=T2	GC-D	0.899	7.139	11.902	24.900
ossl_lucas_block	diag_mode=T2	R-U-D	0.889	6.845	12.009	24.900
ossl_lucas_block	diag_mode=T2	SCP	0.883	6.086	12.855	24.900
ossl_lucas_block	floor_frac=0.02	G-W-A	0.899	6.470	12.496	24.900
ossl_lucas_block	floor_frac=0.02	GC-D	0.898	7.142	11.900	24.900
ossl_lucas_block	floor_frac=0.02	R-U-D	0.889	6.855	11.998	24.900
ossl_lucas_block	floor_frac=0.0	SCP	0.883	6.086	12.855	24.900

task	variant	method	coverage	width	interval_score	A
	2					
ossl_lucas_block	floor_frac=0.3	G-W-A	0.899	6.470	12.496	24.900
ossl_lucas_block	floor_frac=0.3	GC-D	0.898	7.142	11.900	24.900
ossl_lucas_block	floor_frac=0.3	R-U-D	0.889	6.855	11.998	24.900
ossl_lucas_block	floor_frac=0.3	SCP	0.883	6.086	12.855	24.900

S6. Convolutional network as point predictor

Table S6. 1D-CNN point predictor (mango, new season; group-averaged coverage over seven seasons; mean over seeds). σ(x) uses the PLS diagnostics of the same spectra.

method	coverage	coverage_min	width	interval_score	rmse	n_seasons	n_seeds
CNN-G-W-A	0.881	0.664	4.606	6.293	1.403	7	3
CNN-GC-D	0.886	0.706	4.563	6.106	1.403	7	3
CNN-R-U-A	0.646	0.401	2.523	8.483	1.403	7	3
CNN-SCP	0.651	0.442	2.533	8.297	1.393	7	3

S7. Post-hoc iteration (GC-D2, GC-CQR)

Table S7a. Post-hoc variants (seeds 0–2; nominal 0.90).

task	method	coverage	coverage_min	width	interval_score	n_groups
corn_moisture	CQR	0.901	0.817	1.191	1.528	30
corn_moisture	GC-CQR	0.814	0.650	0.959	1.547	30
corn_moisture	GC-D	0.529	0.000	0.879	4.753	30
corn_moisture	GC-D2	0.538	0.000	0.851	4.951	30
corn_moisture	SCP	0.612	0.000	0.994	4.635	30
corn_oil	CQR	0.889	0.683	0.522	0.642	30
corn_oil	GC-CQR	0.877	0.667	0.463	0.581	30
corn_oil	GC-D	0.766	0.317	0.406	0.760	30
corn_oil	GC-D2	0.752	0.317	0.396	0.774	30
corn_oil	SCP	0.810	0.350	0.455	0.855	30
corn_protein	CQR	0.927	0.767	1.310	1.514	30
corn_protein	GC-CQR	0.898	0.633	1.137	1.414	30
corn_protein	GC-D	0.710	0.083	1.098	2.119	30
corn_protein	GC-D2	0.673	0.117	0.967	2.257	30
corn_protein	SCP	0.664	0.000	0.970	2.606	30
corn_starch	CQR	0.836	0.417	2.263	3.395	30
corn_starch	GC-CQR	0.827	0.500	2.031	3.283	30
corn_starch	GC-D	0.691	0.000	2.757	6.116	30
corn_starch	GC-D2	0.677	0.000	2.559	6.017	30
corn_starch	SCP	0.600	0.000	1.983	7.345	30
mango_instrument	CQR	0.770	0.363	3.621	7.030	24
mango_instrument	GC-CQR	0.900	0.459	4.946	6.360	24

task	method	coverage	coverage_min	width	interval_score	n_groups
mango_instrument	GC-D	0.879	0.356	4.542	6.211	24
mango_instrument	GC-D2	0.879	0.312	4.568	6.302	24
mango_instrument	SCP	0.793	0.239	3.647	6.951	24
mango_population	CQR	0.840	0.042	3.362	5.236	199
mango_population	GC-CQR	0.903	0.108	3.879	4.980	199
mango_population	GC-D	0.899	0.013	3.686	4.932	199
mango_population	GC-D2	0.898	0.013	3.665	4.915	199
mango_population	SCP	0.886	0.013	3.538	4.971	199
mango_season_forward	CQR	0.727	0.642	3.470	7.474	5
mango_season_forward	GC-CQR	0.858	0.790	4.560	6.366	5
mango_season_forward	GC-D	0.877	0.772	4.526	6.143	5
mango_season_forward	GC-D2	0.879	0.765	4.532	6.136	5
mango_season_forward	SCP	0.780	0.687	3.458	7.241	5
mango_season_loso	CQR	0.788	0.673	3.532	6.405	7
mango_season_loso	GC-CQR	0.901	0.803	4.665	5.877	7
mango_season_loso	GC-D	0.895	0.798	4.519	5.900	7
mango_season_loso	GC-D2	0.892	0.765	4.509	5.974	7
mango_season_loso	SCP	0.825	0.722	3.632	6.447	7
ossl_lucas_block	CQR	0.886	0.814	5.520	8.038	30
ossl_lucas_block	GC-CQR	0.902	0.834	5.612	7.767	30
ossl_lucas_block	GC-D	0.898	0.786	7.142	11.900	30
ossl_lucas_block	GC-D2	0.898	0.789	10.430	13.529	30
ossl_lucas_block	SCP	0.884	0.760	6.110	12.777	30

Table S7b. Paired Wilcoxon tests over held-out groups (Holm across scenarios for each contrast).

task	method	comparator	n_groups	IS_median_diff	IS_r_rb	IS_p	IS_p_holm
corn_moisture	GC-D2	GC-D	30	-0.0078	0.0151	0.9426	1.0000
corn_moisture	GC-CQR	CQR	30	-0.0408	0.0194	0.9263	0.9263
corn_moisture	GC-CQR	GC-D	30	-0.5161	-0.7634	0.0003	0.0016
corn_oil	GC-D2	GC-D	30	-0.0025	-0.1570	0.4528	1.0000
corn_oil	GC-CQR	CQR	30	-0.0506	-0.6946	0.0009	0.0063
corn_oil	GC-CQR	GC-D	30	-0.1157	-0.7763	0.0002	0.0014
corn_protein	GC-D2	GC-D	30	0.0294	0.4323	0.0387	0.3098
corn_protein	GC-CQR	CQR	30	-0.0987	-0.7333	0.0005	0.0036
corn_protein	GC-CQR	GC-D	30	-0.3574	-0.8237	0.0001	0.0007
corn_starch	GC-D2	GC-D	30	-0.1305	-0.4323	0.0387	0.3098
corn_starch	GC-CQR	CQR	30	-0.1988	-0.2774	0.1846	0.6250
corn_starch	GC-CQR	GC-D	30	-0.8402	-0.7247	0.0005	0.0026
mango_instrument	GC-D2	GC-D	24	0.0035	0.3400	0.1515	0.9090
mango_instrument	GC-CQR	CQR	24	-0.3300	-0.6133	0.0072	0.0431
mango_instrument	GC-CQR	GC-D	24	0.3357	0.4800	0.0395	0.1579
mango_population	GC-D2	GC-D	199	-0.0314	-0.2258	0.0057	0.0516
mango_population	GC-CQR	CQR	199	0.0439	-0.1082	0.1855	0.6250

task	method	comparator	n_groups	IS_median_diff	IS_r_rb	IS_p	IS_p_holm
mango_population	GC-CQR	GC-D	199	0.0676	0.1305	0.1106	0.3317
mango_season_forward	GC-D2	GC-D	5	-0.0145	-0.0667	1.0000	1.0000
mango_season_forward	GC-CQR	CQR	5	-0.9971	-1.0000	0.0625	0.3125
mango_season_forward	GC-CQR	GC-D	5	0.1743	0.7333	0.1875	0.3750
mango_season_loso	GC-D2	GC-D	7	0.0036	0.2143	0.6875	1.0000
mango_season_loso	GC-CQR	CQR	7	-0.7655	-0.6429	0.1562	0.6250
mango_season_loso	GC-CQR	GC-D	7	-0.0328	-0.0714	0.9375	0.9375
ossl_lucas_block	GC-D2	GC-D	30	-0.1455	-0.1484	0.4779	1.0000
ossl_lucas_block	GC-CQR	CQR	30	-0.1405	-0.9828	0.0000	0.0000
ossl_lucas_block	GC-CQR	GC-D	30	-1.6531	-1.0000	0.0000	0.0000

S8. Weighted conformal prediction with clipped density ratios

Table S8. WCP vs WCP-clip (density ratios normalised to mean 1 on the calibration set and capped at 20); finite_frac = share of test spectra with a bounded interval.

task	method	coverage	width_median	interval_score	finite_frac	n_groups	n_seeds
corn_moisture	WCP	0.964	∞	∞	0.535	30	5
corn_moisture	WCP-clip	0.964	∞	∞	0.535	30	5
corn_oil	WCP	0.945	∞	∞	0.502	30	5
corn_oil	WCP-clip	0.945	∞	∞	0.502	30	5
corn_protein	WCP	0.989	∞	∞	0.243	30	5
corn_protein	WCP-clip	0.989	∞	∞	0.243	30	5
corn_starch	WCP	0.936	∞	∞	0.539	30	5
corn_starch	WCP-clip	0.936	∞	∞	0.539	30	5
mango_instrument	WCP	0.905	∞	∞	0.805	24	5
mango_instrument	WCP-clip	0.851	4.106	∞	0.990	24	5
mango_population	WCP	0.907	∞	∞	0.895	199	5
mango_population	WCP-clip	0.899	3.595	∞	1.000	199	5
mango_season_forward	WCP	0.875	10.294	∞	0.999	5	5
mango_season_forward	WCP-clip	0.822	3.664	7.306	1.000	5	5
mango_season_loso	WCP	0.880	3.886	∞	1.000	7	5
mango_season_loso	WCP-clip	0.855	3.946	6.258	1.000	7	5
ossl_kssl_to_lucas	WCP	1.000	∞	∞	0.000	30	5
ossl_kssl_to_lucas	WCP-clip	0.106	31.198	119.848	1.000	30	5
ossl_lucas_block	WCP	0.917	∞	∞	0.980	30	5
ossl_lucas_block	WCP-clip	0.921	4.830	∞	1.000	30	5
ossl_lucas_campaign	WCP	0.803	∞	∞	0.790	58	5
ossl_lucas_campaign	WCP-clip	0.905	∞	∞	0.887	58	5
tablets	WCP	0.962	∞	∞	0.500	2	5
tablets	WCP-clip	0.962	∞	∞	0.500	2	5

S9. Computing cost

Table S9. Isolated computing cost per held-out season (mango LOSO, ~70 000 training spectra, one CPU core, mean of three seasons).

method	method_s	method_s_sd	peak_MiB	ms_per_test_spectrum	shared_setup_s
ASTM-G	0.002	0.000	2.242	0.000	4.895
G-U-A	0.004	0.000	4.575	0.000	4.742
G-W-A	0.012	0.001	5.675	0.001	4.695
ORACLE	0.020	0.011	0.747	0.002	5.190
GC-D+	0.117	0.011	83.338	0.014	4.670
G-U-D	0.120	0.008	83.338	0.015	4.887
GC-D	0.120	0.005	83.338	0.014	4.746
GC-D2	0.146	0.009	82.218	0.017	5.120
HJ+	0.566	0.022	33.700	0.066	4.766
HJ+-D	0.637	0.021	84.985	0.074	4.886
WCP	0.688	0.020	171.217	0.082	5.155
SCP	0.692	0.069	171.217	0.084	5.315
WCP-clip	0.713	0.032	171.217	0.082	5.093
HCP-A	0.771	0.067	132.092	0.092	5.210
HCP-D	1.213	0.132	136.379	0.140	5.085
R-U-A	2.936	0.223	201.935	0.358	4.914
ASTM-R	3.161	0.378	201.935	0.393	5.499
R-U-D	3.236	0.370	201.935	0.400	4.846
CV+	3.603	0.034	201.935	0.421	5.281
CQR	5.213	0.239	171.217	0.613	5.218
QGB	5.257	0.361	171.573	0.615	4.832
NCP-kNN	8.933	0.579	179.120	1.028	5.015
BAG	14.980	2.041	198.978	1.791	4.685
GPR	15.613	4.368	448.503	1.700	5.002
GC-CQR	34.502	2.496	216.729	4.156	4.882

method_s and peak_MiB are measured after the shared step (group-wise cross-validation and final PLS fit, shared_setup_s), which every PLS-based method needs. Sources of the memory differences: ORACLE, ASTM-G, G-U-A and G-W-A only sort residuals that already exist (< 6 MiB); GC-D, G-U-D and GC-D+ evaluate T^2 and Q of every training spectrum under its out-of-group fold model, which requires reconstructing the n × p training matrix in double precision (≈ 70 000 × 103 × 8 bytes ≈ 58 MB, plus the scale-model fit); SCP, WCP and HCP fit an additional PLS model on part of the data; CV+, ASTM-R and R-U-* run a second (random) cross-validation; GPR stores the 1000 × 1000 kernel matrix and its factorisation; bagging stores 25 models.

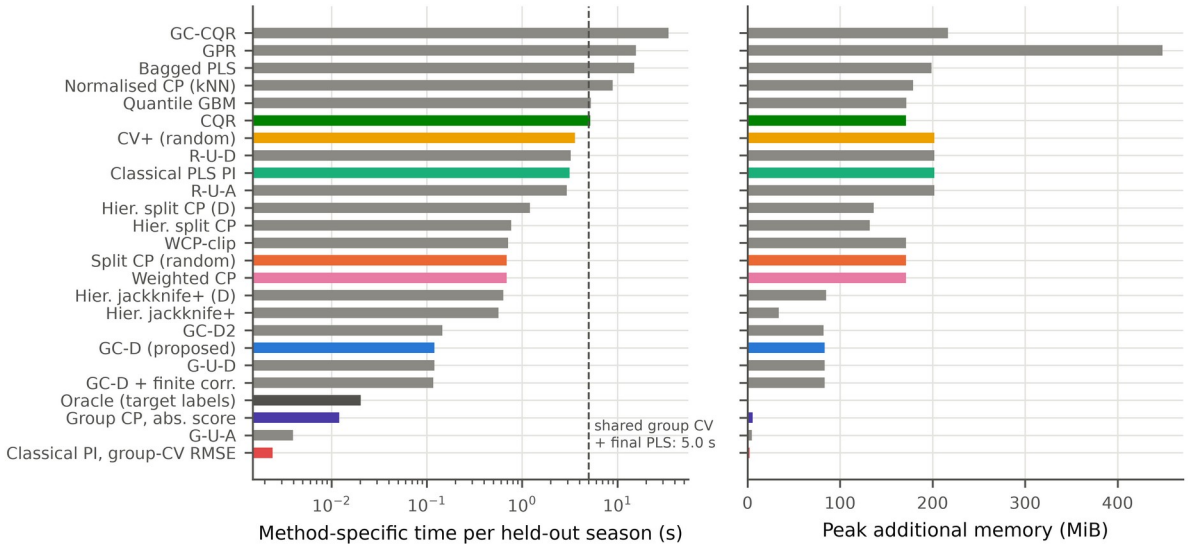


Fig. S1. Method-specific computing time (left, log scale) and peak additional memory (right); dashed line: shared group-wise cross-validation and final PLS fit.

S10. Pooled coverage with cluster-bootstrap confidence intervals

Table S10. Coverage pooled over all test spectra with 95 % cluster-bootstrap intervals (clusters = fruit or soil sample; seed 0).

task	method	pooled_coverage	ci_lo	ci_hi	n_spectra	n_units
mango_season_loso	ASTM-R	0.870	0.864	0.876	85401	10560
mango_season_loso	BAG	0.869	0.863	0.875	85401	10560
mango_season_loso	CQR	0.828	0.822	0.835	85401	10560
mango_season_loso	CV+	0.863	0.857	0.869	85401	10560
mango_season_loso	G-W-A	0.926	0.921	0.931	85401	10560
mango_season_loso	GC-D	0.927	0.923	0.932	85401	10560
mango_season_loso	GPR	0.853	0.847	0.860	85401	10560
mango_season_loso	ORACLE	0.899	0.894	0.904	85401	10560
mango_season_loso	SCP	0.859	0.853	0.865	85401	10560
mango_season_loso	WCP	0.878	0.872	0.884	85401	10560
mango_instrument	ASTM-R	0.836	0.830	0.842	82383	10508
mango_instrument	BAG	0.838	0.832	0.844	82383	10508
mango_instrument	CQR	0.796	0.789	0.803	82383	10508
mango_instrument	CV+	0.828	0.822	0.834	82383	10508
mango_instrument	G-W-A	0.906	0.902	0.911	82383	10508
mango_instrument	GC-D	0.911	0.906	0.915	82383	10508
mango_instrument	GPR	0.785	0.779	0.791	82383	10508
mango_instrument	ORACLE	0.900	0.894	0.905	82383	10508
mango_instrument	SCP	0.827	0.821	0.834	82383	10508
mango_instrument	WCP	0.919	0.914	0.924	82383	10508
mango_population	ASTM-R	0.892	0.886	0.898	85401	10560
mango_population	BAG	0.892	0.886	0.898	85401	10560
mango_population	CQR	0.855	0.848	0.861	85401	10560
mango_population	CV+	0.886	0.880	0.892	85401	10560

task	method	pooled_coverage	ci_lo	ci_hi	n_spectra	n_units
mango_population	G-W-A	0.898	0.893	0.904	85401	10560
mango_population	GC-D	0.896	0.891	0.902	85401	10560
mango_population	GPR	0.875	0.869	0.882	85401	10560
mango_population	ORACLE	0.897	0.892	0.901	85401	10560
mango_population	SCP	0.882	0.876	0.888	85401	10560
mango_population	WCP	0.908	0.902	0.914	85401	10560
ossl_lucas_block	ASTM-R	0.894	0.891	0.897	40175	40175
ossl_lucas_block	BAG	0.894	0.891	0.897	40175	40175
ossl_lucas_block	CQR	0.886	0.883	0.889	40175	40175
ossl_lucas_block	CV+	0.888	0.885	0.891	40175	40175
ossl_lucas_block	G-W-A	0.902	0.900	0.905	40175	40175
ossl_lucas_block	GC-D	0.898	0.895	0.901	40175	40175
ossl_lucas_block	GPR	0.888	0.885	0.891	40175	40175
ossl_lucas_block	ORACLE	0.901	0.898	0.904	40175	40175
ossl_lucas_block	SCP	0.886	0.883	0.889	40175	40175
ossl_lucas_block	WCP	0.883	0.880	0.886	40175	40175
mango_season_forward	ASTM-R	0.828	0.821	0.835	54891	7711
mango_season_forward	BAG	0.832	0.825	0.838	54891	7711
mango_season_forward	CQR	0.769	0.760	0.779	54891	7711
mango_season_forward	CV+	0.820	0.813	0.827	54891	7711
mango_season_forward	G-W-A	0.903	0.897	0.908	54891	7711
mango_season_forward	GC-D	0.920	0.914	0.925	54891	7711
mango_season_forward	GPR	0.799	0.791	0.807	54891	7711
mango_season_forward	ORACLE	0.900	0.895	0.905	54891	7711
mango_season_forward	SCP	0.822	0.815	0.829	54891	7711
mango_season_forward	WCP	0.911	0.904	0.916	54891	7711

S11. Specification-limit decisions

Table S11. Decisions 'DM ≥ L' accepted when the lower interval limit ≥ L (mango, new season, seed 0).

limit	method	n	accepted	false_accept_among_accepted	false_accept_overall	yield_of_compliant
14.000	ASTM-R	85401	0.476	0.022	0.011	0.616
14.000	BAG	85401	0.478	0.022	0.011	0.618
14.000	CQR	85401	0.517	0.027	0.014	0.665
14.000	CV+	85401	0.484	0.023	0.011	0.625
14.000	G-W-A	85401	0.411	0.013	0.006	0.536
14.000	GC-D	85401	0.422	0.013	0.005	0.551
14.000	GPR	85401	0.492	0.023	0.011	0.636
14.000	ORACLE	85401	0.454	0.015	0.007	0.591
14.000	SCP	85401	0.488	0.023	0.011	0.631
14.000	WCP	85401	0.475	0.020	0.009	0.615
15.000	ASTM-R	85401	0.324	0.022	0.007	0.520

limit	method	n	accepted	false_accept_among_accepted	false_accept_overall	yield_of_compliant
15.000	BAG	85401	0.327	0.022	0.007	0.524
15.000	CQR	85401	0.331	0.013	0.004	0.535
15.000	CV+	85401	0.331	0.023	0.008	0.529
15.000	G-W-A	85401	0.271	0.015	0.004	0.438
15.000	GC-D	85401	0.279	0.009	0.003	0.453
15.000	GPR	85401	0.339	0.015	0.005	0.547
15.000	ORACLE	85401	0.301	0.016	0.005	0.485
15.000	SCP	85401	0.335	0.024	0.008	0.536
15.000	WCP	85401	0.319	0.017	0.006	0.513
16.000	ASTM-R	85401	0.204	0.039	0.008	0.416
16.000	BAG	85401	0.205	0.039	0.008	0.419
16.000	CQR	85401	0.168	0.017	0.003	0.351
16.000	CV+	85401	0.208	0.039	0.008	0.426
16.000	G-W-A	85401	0.161	0.030	0.005	0.332
16.000	GC-D	85401	0.164	0.021	0.003	0.342
16.000	GPR	85401	0.214	0.019	0.004	0.446
16.000	ORACLE	85401	0.181	0.026	0.005	0.375
16.000	SCP	85401	0.213	0.041	0.009	0.434
16.000	WCP	85401	0.196	0.030	0.006	0.404
17.000	ASTM-R	85401	0.111	0.056	0.006	0.318
17.000	BAG	85401	0.112	0.055	0.006	0.322
17.000	CQR	85401	0.048	0.007	0.000	0.146
17.000	CV+	85401	0.114	0.056	0.006	0.329
17.000	G-W-A	85401	0.082	0.044	0.004	0.237
17.000	GC-D	85401	0.081	0.024	0.002	0.240
17.000	GPR	85401	0.115	0.034	0.004	0.338
17.000	ORACLE	85401	0.093	0.035	0.003	0.272
17.000	SCP	85401	0.118	0.057	0.007	0.339
17.000	WCP	85401	0.105	0.042	0.004	0.307

S12. Informativeness of the corn models on a new instrument

Table S12. RMSEP and bias on the held-out instrument vs. the SD of the reference values, and CQR width vs. the central 90 % range of the reference values (seed 0).

property	test_instrument	rmsep	bias	sd_reference	cqr_width	marginal_90_range
moisture	m5	0.931	0.893	0.380	1.232	1.185
moisture	mp5	0.345	-0.148	0.380	1.240	1.185
moisture	mp6	0.313	0.018	0.380	1.399	1.185
oil	m5	0.153	-0.071	0.177	0.534	0.567
oil	mp5	0.173	0.047	0.177	0.590	0.567
oil	mp6	0.172	0.067	0.177	0.572	0.567
protein	m5	0.456	0.408	0.499	1.215	1.583

property	test_instrument	rmsep	bias	sd_reference	cqr_width	marginal_90_range
protein	mp5	0.398	0.074	0.499	1.486	1.583
protein	mp6	0.304	-0.126	0.499	1.406	1.583
starch	m5	1.315	-1.263	0.821	2.121	2.557
starch	mp5	0.829	0.121	0.821	2.621	2.557
starch	mp6	0.683	0.186	0.821	2.359	2.557

S13. Number of calibration groups

Table S13. Coverage when only K training groups are used (three repetitions); finite = minimum share of bounded intervals.

task	method	K	coverage	coverage_sd	width	finite
mango_instrument	CV+	3	0.636	0.085	4.016	1.000
mango_instrument	CV+	5	0.675	0.031	4.238	1.000
mango_instrument	CV+	10	0.727	0.052	3.966	1.000
mango_instrument	CV+	20	0.755	0.019	3.791	1.000
mango_instrument	G-W-A	3	0.786	0.078	5.579	1.000
mango_instrument	G-W-A	5	0.800	0.007	5.520	1.000
mango_instrument	G-W-A	10	0.868	0.045	5.433	1.000
mango_instrument	G-W-A	20	0.866	0.013	4.994	1.000
mango_instrument	GC-D	3	0.757	0.068	5.474	1.000
mango_instrument	GC-D	5	0.782	0.019	5.627	1.000
mango_instrument	GC-D	10	0.861	0.030	5.327	1.000
mango_instrument	GC-D	20	0.865	0.011	4.854	1.000
mango_instrument	HCP-D	3	1.000	0.000	∞	0.000
mango_instrument	HCP-D	5	1.000	0.000	∞	0.000
mango_instrument	HCP-D	10	1.000	0.000	∞	0.000
mango_instrument	HCP-D	20	0.972	0.022	17.622	1.000
mango_instrument	HJ+	3	1.000	0.000	∞	0.000
mango_instrument	HJ+	5	1.000	0.000	∞	0.000
mango_instrument	HJ+	10	0.987	0.005	10.153	1.000
mango_instrument	HJ+	20	0.942	0.003	6.142	1.000
mango_instrument	ORACLE	3	0.899	0.001	7.512	1.000
mango_instrument	ORACLE	5	0.900	0.001	8.310	1.000
mango_instrument	ORACLE	10	0.900	0.000	5.405	1.000
mango_instrument	ORACLE	20	0.900	0.001	5.062	1.000
mango_instrument	SCP	3	0.627	0.085	3.981	1.000
mango_instrument	SCP	5	0.672	0.036	4.227	1.000
mango_instrument	SCP	10	0.725	0.049	3.968	1.000
mango_instrument	SCP	20	0.754	0.022	3.779	1.000
ossl_lucas_block	CV+	3	0.805	0.013	1182.591	1.000
ossl_lucas_block	CV+	5	0.834	0.011	75.373	1.000
ossl_lucas_block	CV+	10	0.868	0.013	366.406	1.000
ossl_lucas_block	CV+	20	0.877	0.002	6.112	1.000
ossl_lucas_block	G-W-A	3	0.877	0.016	452.973	1.000

task	method	K	coverage	coverage_sd	width	finite
ossl_lucas_block	G-W-A	5	0.887	0.009	18.926	1.000
ossl_lucas_block	G-W-A	10	0.900	0.010	15.037	1.000
ossl_lucas_block	G-W-A	20	0.898	0.003	6.638	1.000
ossl_lucas_block	GC-D	3	0.864	0.012	7681.947	1.000
ossl_lucas_block	GC-D	5	0.876	0.007	60.569	1.000
ossl_lucas_block	GC-D	10	0.902	0.008	45.105	1.000
ossl_lucas_block	GC-D	20	0.898	0.002	7.453	1.000
ossl_lucas_block	HCP-D	3	1.000	0.000	∞	0.000
ossl_lucas_block	HCP-D	5	1.000	0.000	∞	0.000
ossl_lucas_block	HCP-D	10	1.000	0.000	∞	0.000
ossl_lucas_block	HCP-D	20	0.990	0.002	∞	1.000
ossl_lucas_block	HJ+	3	1.000	0.000	∞	0.000
ossl_lucas_block	HJ+	5	1.000	0.000	∞	0.000
ossl_lucas_block	HJ+	10	0.989	0.001	8892.885	1.000
ossl_lucas_block	HJ+	20	0.945	0.001	8.502	1.000
ossl_lucas_block	ORACLE	3	0.902	0.001	529.612	1.000
ossl_lucas_block	ORACLE	5	0.901	0.001	20.594	1.000
ossl_lucas_block	ORACLE	10	0.902	0.001	17.834	1.000
ossl_lucas_block	ORACLE	20	0.901	0.000	7.274	1.000
ossl_lucas_block	SCP	3	0.803	0.010	14.950	1.000
ossl_lucas_block	SCP	5	0.833	0.012	13.326	1.000
ossl_lucas_block	SCP	10	0.869	0.013	15.605	1.000
ossl_lucas_block	SCP	20	0.876	0.002	6.081	1.000

S14. Paired comparisons

Table S14. All paired comparisons of GC-D with each comparator (two-sided Wilcoxon signed-rank over units; ΔIS = median of GC-D minus comparator interval score, −∞ when the comparator's intervals were unbounded; r = rank-biserial correlation; Holm within scenario and, more conservatively, across all main-scenario tests (all); Δ|cov. gap| = median difference of |coverage − 0.90|).

Scenario	Comparator	Unit	n	ΔIS (median)	r	p	p Holm	p Holm (all)	Δ|cov. gap|	p Holm (gap)
corn_moisture	ASTM-G	group	30	0.0748	0.1398	0.5038	1.0000	1.0000	0.0000	0.2422
corn_moisture	ASTM-R	group	30	0.0859	0.0839	0.6884	1.0000	1.0000	0.0000	0.2422
corn_moisture	BAG	group	30	0.1250	0.3548	0.0897	0.9869	1.0000	0.0150	0.1609
corn_moisture	CQR	group	30	0.8063	0.7677	0.0002	0.0031	0.0340	0.2750	0.0002
corn_moisture	CV+	group	30	0.0882	0.1699	0.4165	1.0000	1.0000	0.0100	0.2321
corn_moisture	G-U-A	group	30	0.0403	0.0280	0.8936	1.0000	1.0000	0.0000	0.2422
corn_moisture	G-U-D	group	30	0.0000	0.0000	1.0000	1.0000	1.0000	0.0000	1.0000
corn_moisture	G-W-A	group	30	0.0403	0.0280	0.8936	1.0000	1.0000	0.0000	0.2422
corn_moisture	GC-D+	group	30	−∞	-1.0000	0.0000	0.0000	0.0003	0.2050	0.0075
corn_moisture	GPR	group	30	0.5050	0.7720	0.0002	0.0031	0.0316	0.0850	0.0021
corn_moisture	HCP-A	group	30	−∞	-1.0000	0.0000	0.0000	0.0003	0.2050	0.0075
corn_moisture	HCP-D	group	30	−∞	-1.0000	0.0000	0.0000	0.0003	0.2050	0.0075
corn_moisture	HJ+	group	30	−∞	-1.0000	0.0000	0.0000	0.0003	0.2050	0.0075

Scenario	Comparator	Unit	n	ΔIS (median)	r	p	p Holm	p Holm (all)	Δ|cov. gap|	p Holm (gap)
corn_moisture	HJ+-D	group	30	−∞	-1.0000	0.0000	0.0000	0.0003	0.2050	0.0075
corn_moisture	NCP-kNN	group	30	-0.0029	0.0452	0.8290	1.0000	1.0000	0.0200	0.1145
corn_moisture	QGB	group	30	0.5840	0.6946	0.0009	0.0107	0.1234	0.1300	0.0248
corn_moisture	R-U-A	group	30	0.0381	-0.0237	0.9099	1.0000	1.0000	0.0050	0.2321
corn_moisture	R-U-D	group	30	0.1197	0.1570	0.4528	1.0000	1.0000	0.0000	0.2321
corn_moisture	SCP	group	30	0.0727	0.3247	0.1204	1.0000	1.0000	0.0200	0.0139
corn_moisture	WCP	group	30	−∞	-0.9871	0.0000	0.0000	0.0004	0.3000	0.0040
corn_oil	ASTM-G	group	30	0.0272	0.3075	0.1414	1.0000	1.0000	0.0600	0.2347
corn_oil	ASTM-R	group	30	0.0053	0.0968	0.6435	1.0000	1.0000	0.0050	1.0000
corn_oil	BAG	group	30	0.0242	0.4194	0.0449	0.5390	1.0000	0.0450	0.2761
corn_oil	CQR	group	30	0.1022	0.6559	0.0017	0.0239	0.2273	0.0850	0.0088
corn_oil	CV+	group	30	0.0314	0.2903	0.1650	1.0000	1.0000	0.0250	1.0000
corn_oil	G-U-A	group	30	0.0009	0.0925	0.6583	1.0000	1.0000	0.0550	0.2700
corn_oil	G-U-D	group	30	0.0000	0.0000	1.0000	1.0000	1.0000	0.0000	1.0000
corn_oil	G-W-A	group	30	0.0009	0.0925	0.6583	1.0000	1.0000	0.0550	0.2700
corn_oil	GC-D+	group	30	−∞	-1.0000	0.0000	0.0000	0.0003	0.0200	0.9765
corn_oil	GPR	group	30	0.0535	0.4022	0.0545	0.5991	1.0000	0.0600	0.3529
corn_oil	HCP-A	group	30	−∞	-1.0000	0.0000	0.0000	0.0003	0.0200	0.9765
corn_oil	HCP-D	group	30	−∞	-1.0000	0.0000	0.0000	0.0003	0.0200	0.9765
corn_oil	HJ+	group	30	−∞	-1.0000	0.0000	0.0000	0.0003	0.0200	0.9765
corn_oil	HJ+-D	group	30	−∞	-1.0000	0.0000	0.0000	0.0003	0.0200	0.9765
corn_oil	NCP-kNN	group	30	-0.0474	-0.3247	0.1204	1.0000	1.0000	0.0100	1.0000
corn_oil	QGB	group	30	0.0113	0.0882	0.6733	1.0000	1.0000	-0.0550	1.0000
corn_oil	R-U-A	group	30	-0.0085	0.0237	0.9099	1.0000	1.0000	0.0000	1.0000
corn_oil	R-U-D	group	30	0.0639	0.4925	0.0185	0.2407	1.0000	0.0150	0.4197
corn_oil	SCP	group	30	-0.0359	-0.2258	0.2802	1.0000	1.0000	0.0350	1.0000
corn_oil	WCP	group	30	−∞	-0.9871	0.0000	0.0000	0.0004	0.0450	0.0604
corn_protein	ASTM-G	group	30	0.1044	0.2817	0.1779	1.0000	1.0000	-0.0100	1.0000
corn_protein	ASTM-R	group	30	0.1780	0.1484	0.4779	1.0000	1.0000	-0.0050	1.0000
corn_protein	BAG	group	30	0.1267	0.2086	0.3185	1.0000	1.0000	0.0050	1.0000
corn_protein	CQR	group	30	0.3458	0.6946	0.0009	0.0125	0.1234	0.1250	0.0014
corn_protein	CV+	group	30	0.1527	0.1054	0.6143	1.0000	1.0000	-0.0050	1.0000
corn_protein	G-U-A	group	30	0.0920	0.0753	0.7189	1.0000	1.0000	-0.0050	1.0000
corn_protein	G-U-D	group	30	0.0000	0.0000	1.0000	1.0000	1.0000	0.0000	1.0000
corn_protein	G-W-A	group	30	0.0920	0.0753	0.7189	1.0000	1.0000	-0.0050	1.0000
corn_protein	GC-D+	group	30	−∞	-1.0000	0.0000	0.0000	0.0003	0.0650	0.0794
corn_protein	GPR	group	30	0.2833	0.6645	0.0015	0.0193	0.1988	0.0250	1.0000
corn_protein	HCP-A	group	30	−∞	-1.0000	0.0000	0.0000	0.0003	0.0650	0.0794
corn_protein	HCP-D	group	30	−∞	-1.0000	0.0000	0.0000	0.0003	0.0650	0.0794
corn_protein	HJ+	group	30	−∞	-1.0000	0.0000	0.0000	0.0003	0.0650	0.0794
corn_protein	HJ+-D	group	30	−∞	-1.0000	0.0000	0.0000	0.0003	0.0650	0.0794
corn_protein	NCP-kNN	group	30	0.0470	-0.0968	0.6435	1.0000	1.0000	0.0000	1.0000
corn_protein	QGB	group	30	0.0569	0.2258	0.2802	1.0000	1.0000	0.0000	1.0000
corn_protein	R-U-A	group	30	0.1778	0.0323	0.8774	1.0000	1.0000	-0.0100	1.0000

Scenario	Comparator	Unit	n	ΔIS (median)	r	p	p Holm	p Holm (all)	Δ|cov. gap|	p Holm (gap)
corn_protein	R-U-D	group	30	0.1248	0.4237	0.0428	0.5132	1.0000	0.0150	1.0000
corn_protein	SCP	group	30	0.0454	-0.0968	0.6435	1.0000	1.0000	0.0000	1.0000
corn_protein	WCP	group	30	−∞	-1.0000	0.0000	0.0000	0.0003	0.0850	0.0103
corn_starch	ASTM-G	group	30	0.4593	0.3806	0.0687	0.7221	1.0000	0.0050	1.0000
corn_starch	ASTM-R	group	30	0.3343	0.1484	0.4779	1.0000	1.0000	0.0000	1.0000
corn_starch	BAG	group	30	0.7404	0.4581	0.0285	0.3418	1.0000	0.0150	1.0000
corn_starch	CQR	group	30	0.7189	0.8151	0.0001	0.0014	0.0142	0.0500	0.0064
corn_starch	CV+	group	30	0.3772	0.1484	0.4779	1.0000	1.0000	0.0100	1.0000
corn_starch	G-U-A	group	30	0.3859	0.1957	0.3493	1.0000	1.0000	0.0000	1.0000
corn_starch	G-U-D	group	30	0.0000	0.0000	1.0000	1.0000	1.0000	0.0000	1.0000
corn_starch	G-W-A	group	30	0.3859	0.1957	0.3493	1.0000	1.0000	0.0000	1.0000
corn_starch	GC-D+	group	30	−∞	-1.0000	0.0000	0.0000	0.0003	-0.0150	1.0000
corn_starch	GPR	group	30	0.4886	0.2602	0.2134	1.0000	1.0000	-0.0050	1.0000
corn_starch	HCP-A	group	30	−∞	-1.0000	0.0000	0.0000	0.0003	-0.0150	1.0000
corn_starch	HCP-D	group	30	−∞	-1.0000	0.0000	0.0000	0.0003	-0.0150	1.0000
corn_starch	HJ+	group	30	−∞	-1.0000	0.0000	0.0000	0.0003	-0.0150	1.0000
corn_starch	HJ+-D	group	30	−∞	-1.0000	0.0000	0.0000	0.0003	-0.0150	1.0000
corn_starch	NCP-kNN	group	30	0.0716	-0.1656	0.4284	1.0000	1.0000	0.0000	1.0000
corn_starch	QGB	group	30	0.2059	0.3849	0.0656	0.7221	1.0000	-0.0550	1.0000
corn_starch	R-U-A	group	30	0.3156	0.0065	0.9754	1.0000	1.0000	0.0050	1.0000
corn_starch	R-U-D	group	30	0.6903	0.6258	0.0028	0.0359	0.3567	0.0150	1.0000
corn_starch	SCP	group	30	-0.0360	-0.2000	0.3389	1.0000	1.0000	-0.0250	1.0000
corn_starch	WCP	group	30	−∞	-0.9957	0.0000	0.0000	0.0003	0.0300	0.0432
mango_instrument	ASTM-G	group	24	-0.1169	-0.3667	0.1208	1.0000	1.0000	-0.0014	1.0000
mango_instrument	ASTM-R	group	24	-0.0729	-0.3067	0.1974	1.0000	1.0000	-0.0124	0.7900
mango_instrument	BAG	group	24	-0.0629	-0.2867	0.2292	1.0000	1.0000	-0.0117	0.7900
mango_instrument	CQR	group	24	-0.8632	-0.6467	0.0043	0.0650	0.5506	-0.0416	0.0866
mango_instrument	CV+	group	24	-0.1091	-0.3867	0.1011	1.0000	1.0000	-0.0175	0.6798
mango_instrument	G-U-A	group	24	-0.0252	-0.2067	0.3902	1.0000	1.0000	0.0029	1.0000
mango_instrument	G-U-D	group	24	-0.0321	0.0000	1.0000	1.0000	1.0000	-0.0014	1.0000
mango_instrument	G-W-A	group	24	-0.0537	-0.2733	0.2522	1.0000	1.0000	0.0021	1.0000
mango_instrument	GC-D+	group	24	-0.1321	-0.2667	0.2643	1.0000	1.0000	-0.0103	1.0000
mango_instrument	GPR	group	24	-0.5475	-0.5467	0.0179	0.2502	1.0000	-0.0500	0.1482
mango_instrument	HCP-A	group	24	-6.7965	-1.0000	0.0000	0.0000	0.0000	-0.0296	1.0000
mango_instrument	HCP-D	group	24	-4.8531	-1.0000	0.0000	0.0000	0.0000	-0.0177	1.0000
mango_instrument	HJ+	group	24	-0.2407	-0.2800	0.2405	1.0000	1.0000	-0.0119	1.0000
mango_instrument	HJ+-D	group	24	-0.1533	-0.3133	0.1875	1.0000	1.0000	-0.0126	1.0000
mango_instrument	NCP-kNN	group	24	-0.6518	-0.8800	0.0000	0.0005	0.0045	-0.0571	0.0427
mango_instrument	QGB	group	24	-1.3881	-0.7333	0.0010	0.0154	0.1309	-0.0889	0.0000
mango_instrument	R-U-A	group	24	-0.1189	-0.3933	0.0951	1.0000	1.0000	-0.0177	0.6798
mango_instrument	R-U-D	group	24	-0.0164	-0.1400	0.5646	1.0000	1.0000	-0.0160	1.0000
mango_instrument	SCP	group	24	-0.1264	-0.3867	0.1011	1.0000	1.0000	-0.0167	0.7900
mango_instrument	WCP	group	24	−∞	-1.0000	0.0000	0.0000	0.0000	0.0064	1.0000
mango_population	ASTM-G	fold (seed	10	-0.0471	-0.5636	0.1309	1.0000	1.0000	-0.0022	1.0000

Scenario	Comparator	Unit	n	ΔIS (median)	r	p	p Holm	p Holm (all)	Δ|cov. gap|	p Holm (gap)
		0)								
mango_population	ASTM-R	fold (seed 0)	10	-0.0477	-0.5636	0.1309	1.0000	1.0000	-0.0052	1.0000
mango_population	BAG	fold (seed 0)	10	-0.0471	-0.5636	0.1309	1.0000	1.0000	-0.0044	1.0000
mango_population	CQR	fold (seed 0)	10	-0.3762	-0.7091	0.0488	0.7324	1.0000	-0.0450	0.2598
mango_population	CV+	fold (seed 0)	10	-0.0464	-0.6727	0.0645	0.7734	1.0000	-0.0069	1.0000
mango_population	G-U-A	fold (seed 0)	10	-0.0454	-0.5636	0.1309	1.0000	1.0000	-0.0028	1.0000
mango_population	G-U-D	fold (seed 0)	10	-0.0011	-0.0545	0.9219	1.0000	1.0000	-0.0010	1.0000
mango_population	G-W-A	fold (seed 0)	10	-0.0449	-0.6364	0.0840	0.9238	1.0000	-0.0037	1.0000
mango_population	GC-D+	fold (seed 0)	10	-0.0004	-0.1273	0.7695	1.0000	1.0000	-0.0009	1.0000
mango_population	GPR	fold (seed 0)	10	0.2715	0.7455	0.0371	0.5938	1.0000	-0.0089	1.0000
mango_population	HCP-A	fold (seed 0)	10	-0.1884	-0.8545	0.0137	0.2324	1.0000	-0.0035	1.0000
mango_population	HCP-D	fold (seed 0)	10	-0.1645	-0.8909	0.0098	0.1758	1.0000	-0.0060	1.0000
mango_population	HJ+	fold (seed 0)	10	-0.0486	-0.4182	0.2754	1.0000	1.0000	-0.0040	1.0000
mango_population	HJ+-D	fold (seed 0)	10	-0.0056	0.1636	0.6953	1.0000	1.0000	-0.0020	1.0000
mango_population	NCP-kNN	fold (seed 0)	10	0.2138	0.4182	0.2754	1.0000	1.0000	-0.0457	0.3516
mango_population	QGB	fold (seed 0)	10	-0.7198	-0.9636	0.0039	0.0742	0.5000	-0.0960	0.0391
mango_population	R-U-A	fold (seed 0)	10	-0.0467	-0.7091	0.0488	0.7324	1.0000	-0.0074	1.0000
mango_population	R-U-D	fold (seed 0)	10	0.0018	0.1273	0.7695	1.0000	1.0000	-0.0041	1.0000
mango_population	SCP	fold (seed 0)	10	-0.0527	-0.7091	0.0488	0.7324	1.0000	-0.0144	0.8301
mango_population	WCP	fold (seed 0)	10	−∞	-1.0000	0.0020	0.0391	0.2559	-0.0032	1.0000
mango_season_forward	ASTM-G	group	5	-0.2464	-0.4667	0.4375	1.0000	1.0000	0.0011	1.0000
mango_season_forward	ASTM-R	group	5	-0.0948	-0.4667	0.4375	1.0000	1.0000	-0.0725	1.0000
mango_season_forward	BAG	group	5	-0.0583	-0.4667	0.4375	1.0000	1.0000	-0.0686	1.0000
mango_season_forward	CQR	group	5	-1.1948	-1.0000	0.0625	1.0000	1.0000	-0.1282	1.0000
mango_season_forward	CV+	group	5	-0.1315	-0.6000	0.3125	1.0000	1.0000	-0.0795	1.0000

Scenario	Comparator	Unit	n	ΔIS (median)	r	p	p Holm	p Holm (all)	Δ|cov. gap|	p Holm (gap)
mango_season_forward	G-U-A	group	5	-0.1772	-0.4667	0.4375	1.0000	1.0000	-0.0059	1.0000
mango_season_forward	G-U-D	group	5	-0.0383	-0.4667	0.4375	1.0000	1.0000	-0.0055	1.0000
mango_season_forward	G-W-A	group	5	-0.1775	-0.4667	0.4375	1.0000	1.0000	0.0004	1.0000
mango_season_forward	GC-D+	group	5	−∞	-1.0000	0.0625	1.0000	1.0000	-0.0331	1.0000
mango_season_forward	GPR	group	5	-0.6755	-1.0000	0.0625	1.0000	1.0000	-0.1180	1.0000
mango_season_forward	HCP-A	group	5	−∞	-1.0000	0.0625	1.0000	1.0000	-0.0331	1.0000
mango_season_forward	HCP-D	group	5	−∞	-1.0000	0.0625	1.0000	1.0000	-0.0331	1.0000
mango_season_forward	HJ+	group	5	−∞	-1.0000	0.0625	1.0000	1.0000	-0.0331	1.0000
mango_season_forward	HJ+-D	group	5	−∞	-1.0000	0.0625	1.0000	1.0000	-0.0331	1.0000
mango_season_forward	NCP-kNN	group	5	-0.6276	-0.6000	0.3125	1.0000	1.0000	-0.0675	1.0000
mango_season_forward	QGB	group	5	-1.9253	-1.0000	0.0625	1.0000	1.0000	-0.1819	1.0000
mango_season_forward	R-U-A	group	5	-0.1418	-0.6000	0.3125	1.0000	1.0000	-0.0828	1.0000
mango_season_forward	R-U-D	group	5	-0.0353	-0.4667	0.4375	1.0000	1.0000	-0.0706	1.0000
mango_season_forward	SCP	group	5	-0.0972	-0.4667	0.4375	1.0000	1.0000	-0.0767	1.0000
mango_season_forward	WCP	group	5	-4.3649	-0.8667	0.1250	1.0000	1.0000	-0.0076	1.0000
mango_season_loso	ASTM-G	group	7	-0.1824	-0.3571	0.4688	1.0000	1.0000	-0.0002	1.0000
mango_season_loso	ASTM-R	group	7	-0.0494	-0.3571	0.4688	1.0000	1.0000	-0.0648	1.0000
mango_season_loso	BAG	group	7	-0.0535	-0.3571	0.4688	1.0000	1.0000	-0.0648	1.0000
mango_season_loso	CQR	group	7	-0.5826	-0.6429	0.1562	1.0000	1.0000	-0.1076	1.0000
mango_season_loso	CV+	group	7	-0.0810	-0.4286	0.3750	1.0000	1.0000	-0.0744	1.0000
mango_season_loso	G-U-A	group	7	0.0544	0.0000	1.0000	1.0000	1.0000	-0.0155	1.0000
mango_season_loso	G-U-D	group	7	-0.0465	-0.2143	0.6875	1.0000	1.0000	-0.0184	1.0000
mango_season_loso	G-W-A	group	7	-0.1016	-0.3571	0.4688	1.0000	1.0000	0.0015	1.0000
mango_season_loso	GC-D+	group	7	−∞	-1.0000	0.0156	0.3125	1.0000	-0.0331	0.6250
mango_season_loso	GPR	group	7	-0.6049	-0.5714	0.2188	1.0000	1.0000	-0.0791	1.0000
mango_season_loso	HCP-A	group	7	−∞	-1.0000	0.0156	0.3125	1.0000	-0.0331	0.6250
mango_season_loso	HCP-D	group	7	−∞	-1.0000	0.0156	0.3125	1.0000	-0.0331	0.6250
mango_season_loso	HJ+	group	7	−∞	-1.0000	0.0156	0.3125	1.0000	-0.0331	0.6250
mango_season_loso	HJ+-D	group	7	−∞	-1.0000	0.0156	0.3125	1.0000	-0.0331	0.6250
mango_season_loso	NCP-kNN	group	7	-0.2430	-0.4286	0.3750	1.0000	1.0000	-0.0706	1.0000
mango_season_loso	QGB	group	7	-1.0478	-0.8571	0.0469	0.7031	1.0000	-0.1675	0.6250
mango_season_loso	R-U-A	group	7	-0.0812	-0.5000	0.2969	1.0000	1.0000	-0.0746	1.0000

Scenario	Comparator	Unit	n	ΔIS (median)	r	p	p Holm	p Holm (all)	Δ|cov. gap|	p Holm (gap)
mango_season_loso	R-U-D	group	7	-0.0495	-0.2857	0.5781	1.0000	1.0000	-0.0479	1.0000
mango_season_loso	SCP	group	7	-0.1138	-0.3571	0.4688	1.0000	1.0000	-0.0772	1.0000
mango_season_loso	WCP	group	7	-0.4367	-0.5000	0.2969	1.0000	1.0000	-0.0110	1.0000
ossl_kssl_to_lucas	ASTM-G	fold	1	-30.7141	-1.0000	1.0000	1.0000	–	-0.5333	1.0000
ossl_kssl_to_lucas	ASTM-R	fold	1	-41.0577	-1.0000	1.0000	1.0000	–	-0.5512	1.0000
ossl_kssl_to_lucas	BAG	fold	1	-41.4047	-1.0000	1.0000	1.0000	–	-0.5493	1.0000
ossl_kssl_to_lucas	CQR	fold	1	-10.0420	-1.0000	1.0000	1.0000	–	-0.2944	1.0000
ossl_kssl_to_lucas	CV+	fold	1	-43.7012	-1.0000	1.0000	1.0000	–	-0.5542	1.0000
ossl_kssl_to_lucas	G-U-A	fold	1	-40.1325	-1.0000	1.0000	1.0000	–	-0.5498	1.0000
ossl_kssl_to_lucas	G-U-D	fold	1	2.2407	1.0000	1.0000	1.0000	–	-0.0537	1.0000
ossl_kssl_to_lucas	G-W-A	fold	1	-33.3959	-1.0000	1.0000	1.0000	–	-0.5384	1.0000
ossl_kssl_to_lucas	GC-D+	fold	1	-3.6857	-1.0000	1.0000	1.0000	–	0.0585	1.0000
ossl_kssl_to_lucas	GPR	fold	1	12.3398	1.0000	1.0000	1.0000	–	0.2205	1.0000
ossl_kssl_to_lucas	HCP-A	fold	1	-22.5906	-1.0000	1.0000	1.0000	–	-0.4366	1.0000
ossl_kssl_to_lucas	HCP-D	fold	1	-5.9800	-1.0000	1.0000	1.0000	–	0.1078	1.0000
ossl_kssl_to_lucas	HJ+	fold	1	-27.1396	-1.0000	1.0000	1.0000	–	-0.5262	1.0000
ossl_kssl_to_lucas	HJ+-D	fold	1	-3.5396	-1.0000	1.0000	1.0000	–	0.0652	1.0000
ossl_kssl_to_lucas	NCP-kNN	fold	1	-28.6438	-1.0000	1.0000	1.0000	–	-0.4900	1.0000
ossl_kssl_to_lucas	QGB	fold	1	-13.5568	-1.0000	1.0000	1.0000	–	-0.3150	1.0000
ossl_kssl_to_lucas	R-U-A	fold	1	-43.7225	-1.0000	1.0000	1.0000	–	-0.5544	1.0000
ossl_kssl_to_lucas	R-U-D	fold	1	3.8305	1.0000	1.0000	1.0000	–	-0.1103	1.0000
ossl_kssl_to_lucas	SCP	fold	1	-52.8624	-1.0000	1.0000	1.0000	–	-0.5580	1.0000
ossl_kssl_to_lucas	WCP	fold	1	−∞	-1.0000	1.0000	1.0000	–	0.1955	1.0000
ossl_lucas_block	ASTM-G	group	30	-0.0701	-0.4624	0.0270	0.1622	1.0000	0.0026	1.0000
ossl_lucas_block	ASTM-R	group	30	-0.1610	-0.7333	0.0005	0.0045	0.0630	0.0060	1.0000
ossl_lucas_block	BAG	group	30	-0.1457	-0.6516	0.0018	0.0147	0.2419	0.0047	1.0000
ossl_lucas_block	CQR	group	30	1.5369	1.0000	0.0000	0.0000	0.0003	0.0406	0.0412
ossl_lucas_block	CV+	group	30	-0.1841	-0.7935	0.0001	0.0021	0.0214	0.0047	1.0000
ossl_lucas_block	G-U-A	group	30	-0.1556	-0.6774	0.0012	0.0108	0.1616	0.0055	1.0000
ossl_lucas_block	G-U-D	group	30	-0.0000	0.2215	0.2895	0.9477	1.0000	0.0000	1.0000
ossl_lucas_block	G-W-A	group	30	-0.1585	-0.6387	0.0023	0.0158	0.2932	0.0037	1.0000
ossl_lucas_block	GC-D+	group	30	-0.1610	-0.0710	0.7343	1.0000	1.0000	-0.0128	1.0000
ossl_lucas_block	GPR	group	30	0.7942	0.9398	0.0000	0.0001	0.0010	0.0124	0.6321
ossl_lucas_block	HCP-A	group	30	-1.2277	-0.7505	0.0003	0.0036	0.0464	-0.0251	1.0000
ossl_lucas_block	HCP-D	group	30	-1.1193	-0.7935	0.0001	0.0021	0.0214	-0.0291	1.0000
ossl_lucas_block	HJ+	group	30	-0.1840	-0.2473	0.2369	0.9477	1.0000	-0.0087	1.0000
ossl_lucas_block	HJ+-D	group	30	-0.1200	-0.1011	0.6288	1.0000	1.0000	-0.0134	1.0000
ossl_lucas_block	NCP-kNN	group	30	0.9220	0.9828	0.0000	0.0000	0.0004	0.0342	0.7326
ossl_lucas_block	QGB	group	30	1.1799	0.9957	0.0000	0.0000	0.0003	-0.0017	1.0000
ossl_lucas_block	R-U-A	group	30	-0.1888	-0.7935	0.0001	0.0021	0.0214	0.0056	1.0000
ossl_lucas_block	R-U-D	group	30	0.0041	-0.2860	0.1714	0.8569	1.0000	0.0041	1.0000
ossl_lucas_block	SCP	group	30	-0.1912	-0.8280	0.0001	0.0011	0.0110	0.0063	1.0000
ossl_lucas_block	WCP	group	30	−∞	-0.9140	0.0000	0.0002	0.0018	-0.0023	1.0000
ossl_lucas_campaign	ASTM-G	fold	2	-0.5270	-1.0000	0.5000	1.0000	–	-0.0091	1.0000

Scenario	Comparator	Unit	n	ΔIS (median)	r	p	p Holm	p Holm (all)	Δ|cov. gap|	p Holm (gap)
ossl_lucas_campaign	ASTM-R	fold	2	-0.8751	-1.0000	0.5000	1.0000	–	-0.0089	1.0000
ossl_lucas_campaign	BAG	fold	2	-0.8964	-1.0000	0.5000	1.0000	–	-0.0082	1.0000
ossl_lucas_campaign	CQR	fold	2	2.4328	1.0000	0.5000	1.0000	–	0.0028	1.0000
ossl_lucas_campaign	CV+	fold	2	-1.0897	-1.0000	0.5000	1.0000	–	-0.0170	1.0000
ossl_lucas_campaign	G-U-A	fold	2	-0.7462	-1.0000	0.5000	1.0000	–	-0.0046	1.0000
ossl_lucas_campaign	G-U-D	fold	2	0.0094	0.3333	1.0000	1.0000	–	0.0059	1.0000
ossl_lucas_campaign	G-W-A	fold	2	-0.6831	-1.0000	0.5000	1.0000	–	-0.0076	1.0000
ossl_lucas_campaign	GC-D+	fold	2	0.3831	1.0000	0.5000	1.0000	–	0.0041	1.0000
ossl_lucas_campaign	GPR	fold	2	-0.2450	-0.3333	1.0000	1.0000	–	-0.0655	1.0000
ossl_lucas_campaign	HCP-A	fold	2	0.0004	0.3333	1.0000	1.0000	–	-0.0136	1.0000
ossl_lucas_campaign	HCP-D	fold	2	-0.4856	-0.3333	1.0000	1.0000	–	-0.0260	1.0000
ossl_lucas_campaign	HJ+	fold	2	0.0808	1.0000	0.5000	1.0000	–	-0.0022	1.0000
ossl_lucas_campaign	HJ+-D	fold	2	0.3615	1.0000	0.5000	1.0000	–	0.0034	1.0000
ossl_lucas_campaign	NCP-kNN	fold	2	0.5545	1.0000	0.5000	1.0000	–	-0.0711	1.0000
ossl_lucas_campaign	QGB	fold	2	1.4254	0.3333	1.0000	1.0000	–	-0.0623	1.0000
ossl_lucas_campaign	R-U-A	fold	2	-1.0909	-1.0000	0.5000	1.0000	–	-0.0179	1.0000
ossl_lucas_campaign	R-U-D	fold	2	-0.0877	-1.0000	0.5000	1.0000	–	0.0064	1.0000
ossl_lucas_campaign	SCP	fold	2	-1.3354	-1.0000	0.5000	1.0000	–	-0.0227	1.0000
ossl_lucas_campaign	WCP	fold	2	−∞	-1.0000	0.5000	1.0000	–	-0.0689	1.0000
tablets	ASTM-G	group	2	-1.1514	-1.0000	0.5000	1.0000	–	-0.0191	1.0000
tablets	ASTM-R	group	2	-1.1514	-1.0000	0.5000	1.0000	–	-0.0191	1.0000
tablets	BAG	group	2	-1.2932	-1.0000	0.5000	1.0000	–	-0.0196	1.0000
tablets	CQR	group	2	-13.5524	-1.0000	0.5000	1.0000	–	-0.0176	1.0000
tablets	CV+	group	2	-0.9594	-1.0000	0.5000	1.0000	–	-0.0352	1.0000
tablets	G-U-A	group	2	-1.3245	-0.3333	1.0000	1.0000	–	-0.0489	1.0000
tablets	G-U-D	group	2	0.0000	0.0000	1.0000	1.0000	–	0.0000	1.0000
tablets	G-W-A	group	2	-1.3245	-0.3333	1.0000	1.0000	–	-0.0489	1.0000
tablets	GC-D+	group	2	-0.0156	-0.3333	1.0000	1.0000	–	-0.0002	1.0000
tablets	GPR	group	2	-3.4028	-1.0000	0.5000	1.0000	–	-0.0830	1.0000
tablets	HCP-A	group	2	-4.9206	-1.0000	0.5000	1.0000	–	-0.0759	1.0000
tablets	HCP-D	group	2	-3.8605	-1.0000	0.5000	1.0000	–	-0.0128	1.0000
tablets	HJ+	group	2	-0.9594	-1.0000	0.5000	1.0000	–	-0.0352	1.0000
tablets	HJ+-D	group	2	-0.0208	-0.3333	1.0000	1.0000	–	-0.0002	1.0000
tablets	NCP-kNN	group	2	-1.1166	-1.0000	0.5000	1.0000	–	-0.0157	1.0000
tablets	QGB	group	2	-13.8898	-1.0000	0.5000	1.0000	–	-0.0578	1.0000
tablets	R-U-A	group	2	-1.3245	-0.3333	1.0000	1.0000	–	-0.0489	1.0000
tablets	R-U-D	group	2	0.0000	0.0000	1.0000	1.0000	–	0.0000	1.0000
tablets	SCP	group	2	-2.4731	-1.0000	0.5000	1.0000	–	-0.0630	1.0000
tablets	WCP	group	2	−∞	-1.0000	0.5000	1.0000	–	-0.0385	1.0000

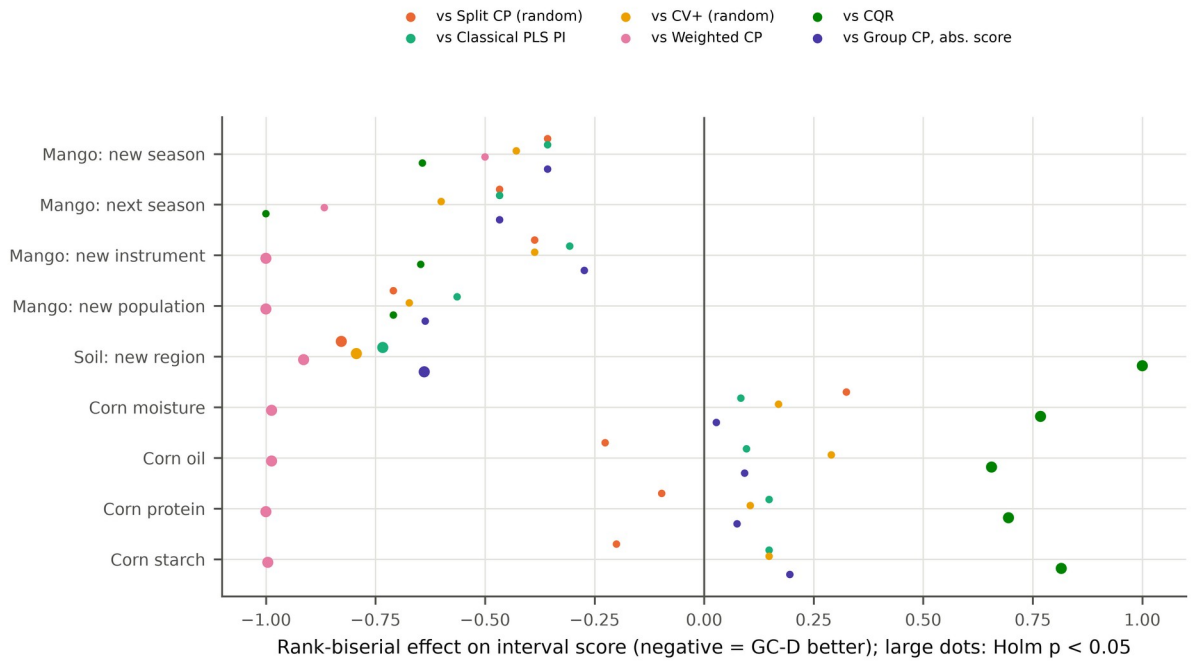


Fig. S2. Rank-biserial effect sizes of the interval-score differences between GC-D and selected comparators (negative: GC-D lower); larger symbols: Holm-adjusted p < 0.05.

S15. Leave-one-cultivar-out test

Table S15a. Leave-one-cultivar-out (mango; ten cultivars; seeds 0–2; nominal 0.90).

method	coverage	coverage_min	width	interval_score	n_cultivars	n_seeds
ASTM-G	0.907	0.805	3.593	4.699	10	3
ASTM-R	0.907	0.788	3.589	4.703	10	3
CQR	0.855	0.738	3.604	5.367	10	3
CV+	0.902	0.776	3.524	4.704	10	3
G-W-A	0.901	0.786	3.509	4.710	10	3
GC-D	0.902	0.778	3.541	4.679	10	3
HJ+	1.000	1.000	20.604	20.604	10	3
ORACLE	0.898	0.892	3.419	4.478	10	3
SCP	0.900	0.772	3.504	4.712	10	3

Table S15b. Paired comparisons of GC-D with each comparator over cultivars (two-sided Wilcoxon, Holm correction).

comparator	n	IS_median_diff	IS_r_rb	IS_p	covgap_median_diff	covgap_p	IS_p_holm	covgap_p_holm
ASTM-G	10	-0.0807	-0.1636	0.6953	0.0073	0.1934	1.0000	0.9668
ASTM-R	10	-0.0722	-0.1273	0.7695	0.0103	0.1055	1.0000	0.6328
CQR	10	-0.6484	-0.8182	0.0195	-0.0064	0.6250	0.1172	1.0000
CV+	10	-0.0796	-0.3091	0.4316	0.0071	0.2324	1.0000	0.9668
G-W-A	10	-0.0997	-0.1636	0.6953	0.0008	0.4922	1.0000	1.0000
HJ+	10	-16.5582	-1.0000	0.0020	-0.0430	0.0098	0.0137	0.0684
SCP	10	-0.0699	-0.3455	0.3750	0.0027	0.4316	1.0000	1.0000

Sections S16–S28: analyses added during revision

The analyses in this part were added in response to peer review, after the main results were known (manuscript Section 3.5). They are produced by experiments/run_revision.py, experiments/run_simulation.py and experiments/revision_analyses.py.

S16. Groups and physical units per scenario

Table S16. Held-out groups, spectra, physical units, units per group (median, min, max), units that occur in more than one group, and the range of the number of calibration groups per outer fold (seed 0).

task	held_out_groups	spectra	units	units_per_group_median	units_per_group_min	units_per_group_max	units_in_more_than_one_group	share_units_shared	calibration_groups_min	calibration_groups_max
mango_season_loso	7	85401	10560	1152.000	562	2894	0	0.000	6	6
mango_season_forward	5	54891	7711	1073.000	562	2894	0	0.000	2	6
mango_instrument	24	82383	10508	1213.500	353	6392	6992	0.665	27	30
mango_population	199	85401	10560	40.000	10	414	0	0.000	179	180
ossl_lucas_block	30	40175	40175	1361.000	76	2650	0	0.000	29	29
corn_moisture	27	219	73	5.000	2	20	73	1.000	2	2
corn_oil	27	219	73	5.000	2	20	73	1.000	2	2
corn_protein	27	219	73	5.000	2	20	73	1.000	2	2
corn_starch	27	219	73	5.000	2	20	73	1.000	2	2
ossl_lucas_campaign	58	40175	40175	730.000	60	1543	0	0.000	28	30
ossl_kssl_to_lucas	30	40175	40175	1361.000	76	2650	0	0.000	84	84
tablets	2	920	460	460.000	460	460	460	1.000	195	195

S17. Form of the scale model σ(x)

Table S17. GC-D with alternative scale models (seed 0; group-averaged coverage, interval score and width at nominal 0.90). Families: Poisson quasi-likelihood (default), Gamma, Gaussian with log link, least squares on log|r| (log-normal); features: log T^2 instead of log(1 + T^2), Q not divided by $\tilde{Q}$, an added $T^2 \times Q$ product term; a fit weighted by the group-balanced weights; and either diagnostic alone.

task	variant	coverage	interval_score	width	n_groups
mango_instrument	Gamma GLM	0.879	6.214	4.532	24
mango_instrument	Gaussian GLM, log link	0.878	6.227	4.563	24
mango_instrument	Q not normalised by $\tilde{Q}$	0.879	6.211	4.542	24
mango_instrument	Q only	0.871	6.326	4.497	24
mango_instrument	T^2 only	0.883	6.220	4.551	24
mango_instrument	default (Poisson, log(1+T^2), Q/$\tilde{Q}$)	0.879	6.211	4.542	24
mango_instrument	group-weighted GLM fit	0.883	6.217	4.616	24
mango_instrument	log T^2 instead of log(1+T^2)	0.878	6.215	4.540	24

task	variant	coverage	interval_score	width	n_groups
mango_instrument	log-normal (LS on log|r|)	0.878	6.211	4.517	24
mango_instrument	with $T^2 \times Q$ interaction	0.880	6.131	4.546	24
mango_season_forward	Gamma GLM	0.874	6.250	4.488	5
mango_season_forward	Gaussian GLM, log link	0.883	6.127	4.643	5
mango_season_forward	Q not normalised by $\tilde{Q}$	0.877	6.143	4.527	5
mango_season_forward	Q only	0.882	6.256	4.611	5
mango_season_forward	T^2 only	0.864	6.343	4.386	5
mango_season_forward	default (Poisson, $\log(1+T^2)$, $Q/\tilde{Q}$)	0.877	6.143	4.526	5
mango_season_forward	group-weighted GLM fit	0.877	6.145	4.533	5
mango_season_forward	log T^2 instead of $\log(1+T^2)$	0.877	6.148	4.528	5
mango_season_forward	log-normal (LS on log|r|)	0.874	6.266	4.503	5
mango_season_forward	with $T^2 \times Q$ interaction	0.874	6.176	4.492	5
mango_season_loso	Gamma GLM	0.896	5.904	4.508	7
mango_season_loso	Gaussian GLM, log link	0.895	5.923	4.557	7
mango_season_loso	Q not normalised by $\tilde{Q}$	0.895	5.899	4.519	7
mango_season_loso	Q only	0.898	6.003	4.561	7
mango_season_loso	T^2 only	0.894	5.939	4.512	7
mango_season_loso	default (Poisson, $\log(1+T^2)$, $Q/\tilde{Q}$)	0.895	5.900	4.519	7
mango_season_loso	group-weighted GLM fit	0.894	6.012	4.515	7
mango_season_loso	log T^2 instead of $\log(1+T^2)$	0.895	5.907	4.518	7
mango_season_loso	log-normal (LS on log|r|)	0.895	5.906	4.499	7
mango_season_loso	with $T^2 \times Q$ interaction	0.900	5.809	4.542	7
ossl_lucas_block	Gamma GLM	0.898	11.913	7.082	30
ossl_lucas_block	Gaussian GLM, log link	0.899	11.893	7.193	30
ossl_lucas_block	Q not normalised by $\tilde{Q}$	0.898	11.900	7.142	30
ossl_lucas_block	Q only	0.898	11.979	6.951	30
ossl_lucas_block	T^2 only	0.899	11.902	7.139	30
ossl_lucas_block	default (Poisson, $\log(1+T^2)$, $Q/\tilde{Q}$)	0.898	11.900	7.142	30
ossl_lucas_block	group-weighted GLM fit	0.898	11.902	7.198	30

task	variant	coverage	interval_score	width	n_groups
ossl_lucas_block	log T^2 instead of $\log(1+T^2)$	0.898	11.902	7.133	30
ossl_lucas_block	log-normal (LS on log|r|)	0.897	11.936	7.032	30
ossl_lucas_block	with $T^2 \times Q$ interaction	0.898	11.932	7.145	30

S18. Strongly unequal group sizes

Table S18. Training groups subsampled to Zipf-like sizes (relative sizes 1, 2^−1.5, 3^−1.5, … in random order; three repetitions). largest_share: median share of the training spectra in the largest group; max_min_ratio: median ratio of the largest to the smallest group. Means over held-out groups; for the log-transformed soil OC the mean interval score and width are dominated by a few back-transformed extreme intervals, so medians over groups are also given.

task	method	coverage	interval_score	interval_score_median	width	width_median	largest_share	max_min_ratio	n_train
mango_instrument	G-U-A	0.831	7.312	5.631	4.373	4.317	0.422	374.688	4294.500
mango_instrument	G-U-D	0.843	6.917	5.852	4.501	4.335	0.422	374.688	4294.500
mango_instrument	G-W-A	0.854	7.111	5.637	4.659	4.694	0.422	374.688	4294.500
mango_instrument	GC-D	0.866	6.742	5.716	4.770	4.622	0.422	374.688	4294.500
mango_instrument	ORACLE	0.900	5.910	5.499	4.774	4.267	0.422	374.688	4294.500
mango_instrument	SCP	0.740	8.103	5.791	3.486	3.480	0.422	374.688	4294.500
mango_season_loso	G-U-A	0.871	6.571	6.129	4.377	4.404	0.590	45.238	21700.000
mango_season_loso	G-U-D	0.854	6.471	6.317	4.261	4.258	0.590	45.238	21700.000
mango_season_loso	G-W-A	0.887	6.522	6.155	4.615	4.545	0.590	45.238	21700.000
mango_season_loso	GC-D	0.876	6.458	6.224	4.571	4.491	0.590	45.238	21700.000
mango_season_loso	ORACLE	0.899	6.223	6.039	4.749	4.729	0.590	45.238	21700.000
mango_season_loso	SCP	0.799	6.951	6.555	3.617	3.540	0.590	45.238	21700.000
ossl_lucas_block	G-U-A	0.886	145.062	7.008	20.584	4.525	0.471	291.300	3130.000
ossl_lucas_block	G-U-D	0.888	1046.222	6.958	1038.296	4.860	0.471	291.300	3130.000
ossl_lucas_block	G-W-A	0.896	143.052	6.936	21.339	4.700	0.471	291.300	3130.000
ossl_lucas_block	GC-D	0.897	921.806	7.122	913.821	4.914	0.471	291.300	3130.000
ossl_lucas_block	ORACLE	0.901	137.856	7.233	23.354	4.373	0.471	291.300	3130.000
ossl_lucas_block	SCP	0.848	66.698	7.154	9.671	3.998	0.471	291.300	3130.000

S19. Interval-score differences: estimation

Table S19. Mean difference in interval score (GC-D − comparator) over units of analysis with 95 % percentile bootstrap interval (4000 resamples), the relative difference, the share of units in which GC-D was lower, and the posterior probability that the mean difference is negative under the Bayesian bootstrap (flat Dirichlet weights on the units). Comparators with unbounded intervals are omitted.

task	comparator	n_units	mean_diff	ci_lo	ci_hi	rel_mean_diff	share_units_gcd_lower	post_prob_gcd_lower
mango_season_loso	SCP	7	-0.534	-1.397	0.143	-0.050	0.571	0.957
mango_season_loso	ASTM-R	7	-0.485	-1.291	0.119	-0.047	0.571	0.960

task	comparator	n_units	mean_diff	ci_lo	ci_hi	rel_mean_diff	share_units_gcd_lower	post_prob_gcd_lower
mango_season_loso	ASTM-G	7	-0.285	-0.876	0.227	-0.033	0.571	0.877
mango_season_loso	G-W-A	7	-0.257	-0.881	0.216	-0.027	0.571	0.858
mango_season_loso	CV+	7	-0.524	-1.299	0.128	-0.051	0.714	0.964
mango_season_loso	CQR	7	-0.522	-1.026	-0.098	-0.068	0.714	0.998
mango_season_forward	SCP	5	-1.066	-2.457	0.255	-0.094	0.600	0.970
mango_season_forward	ASTM-R	5	-1.034	-2.427	0.245	-0.091	0.600	0.973
mango_season_forward	ASTM-G	5	-0.585	-1.493	0.286	-0.062	0.600	0.917
mango_season_forward	G-W-A	5	-0.683	-1.662	0.288	-0.070	0.600	0.937
mango_season_forward	CV+	5	-1.072	-2.564	0.196	-0.096	0.800	0.975
mango_season_forward	CQR	5	-1.349	-1.572	-1.173	-0.180	1.000	1.000
mango_instrument	SCP	24	-0.740	-1.505	-0.150	-0.047	0.667	1.000
mango_instrument	ASTM-R	24	-0.667	-1.395	-0.092	-0.040	0.583	1.000
mango_instrument	ASTM-G	24	-0.268	-0.644	0.022	-0.027	0.708	0.978
mango_instrument	G-W-A	24	-0.265	-0.655	0.046	-0.022	0.625	0.971
mango_instrument	CV+	24	-0.676	-1.400	-0.114	-0.042	0.625	1.000
mango_instrument	CQR	24	-0.865	-1.659	-0.057	-0.084	0.792	0.986
mango_population	SCP	10	-0.057	-0.102	-0.008	-0.011	0.900	0.987
mango_population	ASTM-R	10	-0.037	-0.073	0.002	-0.007	0.700	0.967
mango_population	ASTM-G	10	-0.040	-0.083	0.005	-0.009	0.700	0.963
mango_population	G-W-A	10	-0.040	-0.077	0.003	-0.008	0.700	0.976
mango_population	CV+	10	-0.043	-0.082	-0.000	-0.008	0.900	0.971
mango_population	CQR	10	-0.329	-0.555	-0.097	-0.060	0.700	0.997
ossl_lucas_block	SCP	30	-0.801	-1.619	-0.222	-0.032	0.800	1.000
ossl_lucas_block	ASTM-R	30	-0.732	-1.502	-0.171	-0.028	0.767	1.000
ossl_lucas_block	ASTM-G	30	-0.486	-1.110	-0.024	-0.023	0.700	0.998
ossl_lucas_block	G-W-A	30	-0.595	-1.307	-0.093	-0.025	0.767	1.000
ossl_lucas_block	CV+	30	-0.805	-1.638	-0.229	-0.030	0.800	1.000
ossl_lucas_block	CQR	30	3.844	2.079	5.935	0.350	0.000	0.000
corn_moisture	SCP	30	0.118	-0.333	0.522	0.164	0.333	0.264
corn_moisture	ASTM-R	30	-0.115	-0.550	0.290	0.161	0.433	0.692
corn_moisture	ASTM-G	30	-0.007	-0.413	0.372	0.166	0.433	0.502
corn_moisture	G-W-A	30	-0.273	-0.788	0.180	0.156	0.400	0.867
corn_moisture	CV+	30	-0.021	-0.485	0.394	0.173	0.367	0.514
corn_moisture	CQR	30	3.199	1.669	4.841	2.307	0.267	0.000
corn_oil	SCP	30	-0.099	-0.251	0.015	-0.028	0.533	0.970
corn_oil	ASTM-R	30	-0.012	-0.088	0.059	0.028	0.500	0.598
corn_oil	ASTM-G	30	0.039	-0.020	0.100	0.060	0.400	0.084
corn_oil	G-W-A	30	0.009	-0.061	0.078	0.047	0.500	0.384
corn_oil	CV+	30	0.029	-0.037	0.091	0.054	0.433	0.172

task	comparator	n_units	mean_diff	ci_lo	ci_hi	rel_mean_diff	share_units_gcd_lower	post_prob_gcd_lower
corn_oil	CQR	30	0.172	0.072	0.298	0.256	0.267	0.000
corn_protein	SCP	30	-0.457	-1.067	0.152	0.063	0.467	0.929
corn_protein	ASTM-R	30	0.116	-0.246	0.493	0.198	0.367	0.278
corn_protein	ASTM-G	30	0.195	-0.111	0.532	0.178	0.300	0.102
corn_protein	G-W-A	30	-0.011	-0.428	0.386	0.133	0.367	0.525
corn_protein	CV+	30	0.034	-0.380	0.451	0.174	0.400	0.435
corn_protein	CQR	30	0.621	0.313	0.982	0.401	0.233	0.000
corn_starch	SCP	30	-1.134	-2.202	-0.181	0.015	0.533	0.997
corn_starch	ASTM-R	30	-0.284	-1.157	0.480	0.110	0.300	0.739
corn_starch	ASTM-G	30	0.159	-0.623	0.850	0.121	0.233	0.298
corn_starch	G-W-A	30	-0.333	-1.282	0.440	0.089	0.300	0.778
corn_starch	CV+	30	-0.245	-0.962	0.391	0.106	0.300	0.747
corn_starch	CQR	30	2.744	1.395	4.233	0.877	0.133	0.000

S20. Weighted conformal prediction: density-ratio estimator and cap

Table S20. WCP with three density-ratio estimators (logistic classifier with C = 1 on all PLS scores and log Q, as in the main runs; with C = 0.01; with C = 1 on the first three scores only) and caps on the normalised ratios (seed 0). finite_frac: share of bounded intervals; ess_frac: effective sample size $(\Sigma w)^2/\Sigma w^2$ of the calibration weights relative to the number of calibration spectra (median over groups); test_mass_gt_alpha: share of deployment spectra whose own weight exceeds α of the total; clf_auc: area under the ROC curve of the calibration-vs-deployment classifier.

task	estimator	cap	coverage	finite_frac	interval_score	ess_frac	test_mass_gt_alpha	clf_auc
mango_instrument	logistic, C = 0.01 (strong regularisation)	10.0	0.820	1.000	6.679	0.125	0.000	0.988
mango_instrument	logistic, C = 0.01 (strong regularisation)	100.0	0.858	1.000	6.772	0.040	0.000	0.988
mango_instrument	logistic, C = 0.01 (strong regularisation)	20.0	0.831	1.000	6.678	0.089	0.000	0.988
mango_instrument	logistic, C = 0.01 (strong regularisation)	5.0	0.808	1.000	6.764	0.175	0.000	0.988
mango_instrument	logistic, C = 0.01 (strong regularisation)	50.0	0.848	1.000	6.770	0.056	0.000	0.988
mango_instrument	logistic, C = 0.01 (strong regularisation)	none	0.893	0.927	∞	0.017	0.073	0.988

task	estimator	cap	coverage	finite_frac	interval_score	ess_frac	test_mass_gt_alpha	clf_auc
mango_instrument	logistic, C = 1, all scores + log Q (as in main runs)	10.0	0.848	1.000	7.023	0.036	0.000	0.991
mango_instrument	logistic, C = 1, all scores + log Q (as in main runs)	100.0	0.872	0.945	∞	0.011	0.055	0.991
mango_instrument	logistic, C = 1, all scores + log Q (as in main runs)	20.0	0.855	1.000	7.005	0.025	0.000	0.991
mango_instrument	logistic, C = 1, all scores + log Q (as in main runs)	5.0	0.841	1.000	7.063	0.050	0.000	0.991
mango_instrument	logistic, C = 1, all scores + log Q (as in main runs)	50.0	0.866	0.967	∞	0.015	0.033	0.991
mango_instrument	logistic, C = 1, all scores + log Q (as in main runs)	none	0.926	0.802	∞	0.002	0.198	0.991
mango_instrument	logistic, C = 1, first 3 scores only	10.0	0.792	1.000	6.981	0.441	0.000	0.767
mango_instrument	logistic, C = 1, first 3 scores only	100.0	0.794	1.000	6.972	0.438	0.000	0.767
mango_instrument	logistic, C = 1, first 3 scores only	20.0	0.792	1.000	6.978	0.438	0.000	0.767
mango_instrument	logistic, C = 1, first 3 scores only	5.0	0.791	1.000	6.972	0.484	0.000	0.767
mango_instrument	logistic, C = 1, first 3 scores only	50.0	0.794	1.000	6.972	0.438	0.000	0.767
mango_instrument	logistic, C = 1, first 3 scores only	none	0.800	1.000	7.081	0.438	0.000	0.767
mango_season_forward	logistic, C = 0.01 (strong regularisation)	10.0	0.824	1.000	7.151	0.275	0.000	0.838

task	estimator	cap	coverage	finite_frac	interval_score	ess_frac	test_mass_gt_alpha	clf_auc
mango_season_forward	logistic, C = 0.01 (strong regularisation)	100.0	0.866	1.000	8.348	0.155	0.000	0.838
mango_season_forward	logistic, C = 0.01 (strong regularisation)	20.0	0.838	1.000	8.585	0.199	0.000	0.838
mango_season_forward	logistic, C = 0.01 (strong regularisation)	5.0	0.809	1.000	7.107	0.371	0.000	0.838
mango_season_forward	logistic, C = 0.01 (strong regularisation)	50.0	0.848	1.000	8.478	0.165	0.000	0.838
mango_season_forward	logistic, C = 0.01 (strong regularisation)	none	0.890	1.000	∞	0.016	0.000	0.838
mango_season_forward	logistic, C = 1, all scores + log Q (as in main runs)	10.0	0.833	1.000	∞	0.255	0.000	0.838
mango_season_forward	logistic, C = 1, all scores + log Q (as in main runs)	100.0	0.867	1.000	8.353	0.132	0.000	0.838
mango_season_forward	logistic, C = 1, all scores + log Q (as in main runs)	20.0	0.839	1.000	8.582	0.180	0.000	0.838
mango_season_forward	logistic, C = 1, all scores + log Q (as in main runs)	5.0	0.827	1.000	∞	0.345	0.000	0.838
mango_season_forward	logistic, C = 1, all scores + log Q (as in main runs)	50.0	0.848	1.000	8.478	0.139	0.000	0.838
mango_season_forward	logistic, C = 1, all scores + log Q (as in main runs)	none	0.895	0.999	∞	0.013	0.001	0.838
mango_season_forward	logistic, C = 1, first 3	10.0	0.766	1.000	7.394	0.538	0.000	0.699

task	estimator	cap	coverage	finite_frac	interval_score	ess_frac	test_mass_gt_alpha	clf_auc
	scores only							
mango_season_forward	logistic, C = 1, first 3 scores only	100.0	0.766	1.000	7.395	0.537	0.000	0.699
mango_season_forward	logistic, C = 1, first 3 scores only	20.0	0.766	1.000	7.395	0.537	0.000	0.699
mango_season_forward	logistic, C = 1, first 3 scores only	5.0	0.767	1.000	7.389	0.569	0.000	0.699
mango_season_forward	logistic, C = 1, first 3 scores only	50.0	0.766	1.000	7.395	0.537	0.000	0.699
mango_season_forward	logistic, C = 1, first 3 scores only	none	0.766	1.000	7.395	0.537	0.000	0.699
mango_season_loso	logistic, C = 0.01 (strong regularisation)	10.0	0.840	1.000	6.270	0.320	0.000	0.823
mango_season_loso	logistic, C = 0.01 (strong regularisation)	100.0	0.849	1.000	6.201	0.168	0.000	0.823
mango_season_loso	logistic, C = 0.01 (strong regularisation)	20.0	0.847	1.000	6.230	0.252	0.000	0.823
mango_season_loso	logistic, C = 0.01 (strong regularisation)	5.0	0.832	1.000	6.300	0.396	0.000	0.823
mango_season_loso	logistic, C = 0.01 (strong regularisation)	50.0	0.848	1.000	6.211	0.198	0.000	0.823
mango_season_loso	logistic, C = 0.01 (strong regularisation)	none	0.859	1.000	∞	0.105	0.000	0.823
mango_season_loso	logistic, C = 1, all scores + log Q (as in main runs)	10.0	0.841	1.000	6.267	0.274	0.000	0.825
mango_season_loso	logistic, C = 1, all scores + log Q (as in main runs)	100.0	0.850	1.000	6.197	0.144	0.000	0.825
mango_season_loso	logistic, C = 1, all scores	20.0	0.847	1.000	6.227	0.221	0.000	0.825

task	estimator	cap	coverage	finite_frac	interval_score	ess_frac	test_mass_gt_alpha	clf_auc
	+ log Q (as in main runs)							
mango_season_loso	logistic, C = 1, all scores + log Q (as in main runs)	5.0	0.833	1.000	6.294	0.353	0.000	0.825
mango_season_loso	logistic, C = 1, all scores + log Q (as in main runs)	50.0	0.850	1.000	6.204	0.169	0.000	0.825
mango_season_loso	logistic, C = 1, all scores + log Q (as in main runs)	none	0.864	1.000	∞	0.091	0.000	0.825
mango_season_loso	logistic, C = 1, first 3 scores only	10.0	0.817	1.000	6.455	0.535	0.000	0.707
mango_season_loso	logistic, C = 1, first 3 scores only	100.0	0.817	1.000	6.455	0.532	0.000	0.707
mango_season_loso	logistic, C = 1, first 3 scores only	20.0	0.817	1.000	6.455	0.532	0.000	0.707
mango_season_loso	logistic, C = 1, first 3 scores only	5.0	0.817	1.000	6.454	0.565	0.000	0.707
mango_season_loso	logistic, C = 1, first 3 scores only	50.0	0.817	1.000	6.455	0.532	0.000	0.707
mango_season_loso	logistic, C = 1, first 3 scores only	none	0.817	1.000	6.455	0.532	0.000	0.707
ossl_lucas_block	logistic, C = 0.01 (strong regularisation)	10.0	0.914	1.000	∞	0.183	0.000	0.909
ossl_lucas_block	logistic, C = 0.01 (strong regularisation)	100.0	0.929	1.000	∞	0.060	0.000	0.909
ossl_lucas_block	logistic, C = 0.01 (strong regularisation)	20.0	0.919	1.000	∞	0.128	0.000	0.909
ossl_lucas_block	logistic, C = 0.01 (strong regularisation)	5.0	0.907	1.000	∞	0.260	0.000	0.909

task	estimator	cap	coverage	finite_frac	interval_score	ess_frac	test_mass_gt_alpha	clf_auc
ossl_lucas_block	logistic, C = 0.01 (strong regularisation)	50.0	0.924	1.000	∞	0.081	0.000	0.909
ossl_lucas_block	logistic, C = 0.01 (strong regularisation)	none	0.902	0.996	∞	0.006	0.004	0.909
ossl_lucas_block	logistic, C = 1, all scores + log Q (as in main runs)	10.0	0.919	1.000	∞	0.101	0.000	0.914
ossl_lucas_block	logistic, C = 1, all scores + log Q (as in main runs)	100.0	0.923	0.999	∞	0.022	0.001	0.914
ossl_lucas_block	logistic, C = 1, all scores + log Q (as in main runs)	20.0	0.922	1.000	∞	0.068	0.000	0.914
ossl_lucas_block	logistic, C = 1, all scores + log Q (as in main runs)	5.0	0.913	1.000	∞	0.146	0.000	0.914
ossl_lucas_block	logistic, C = 1, all scores + log Q (as in main runs)	50.0	0.926	1.000	∞	0.037	0.000	0.914
ossl_lucas_block	logistic, C = 1, all scores + log Q (as in main runs)	none	0.891	0.978	∞	0.002	0.022	0.914
ossl_lucas_block	logistic, C = 1, first 3 scores only	10.0	0.896	1.000	11.980	0.528	0.000	0.733
ossl_lucas_block	logistic, C = 1, first 3 scores only	100.0	0.904	1.000	11.940	0.478	0.000	0.733
ossl_lucas_block	logistic, C = 1, first 3 scores only	20.0	0.900	1.000	11.957	0.498	0.000	0.733
ossl_lucas_block	logistic, C = 1, first 3 scores only	5.0	0.893	1.000	12.013	0.575	0.000	0.733
ossl_lucas_block	logistic, C = 1, first 3	50.0	0.902	1.000	11.945	0.478	0.000	0.733

task	estimator	cap	coverage	finite_frac	interval_score	ess_frac	test_mass_gt_alpha	clf_auc
ossl_lucas_block	scores only logistic, C = 1, first 3 scores only	none	0.906	1.000	11.941	0.478	0.000	0.733

S21. Coverage by decile of the Q ratio and of T^2

Table S21. Coverage of SCP, ASTM-G, G-W-A and GC-D by decile of Q/Q̃ and of T^2 of the test spectra (deciles defined on all test spectra of the scenario; seed 0; widths in results/revision/conditional_coverage.csv).

task	by	decile	ASTM-G	G-W-A	GC-D	SCP
mango_instrument	Q/Q̃	1	0.961	0.958	0.907	0.909
mango_instrument	Q/Q̃	2	0.957	0.952	0.909	0.887
mango_instrument	Q/Q̃	3	0.950	0.946	0.908	0.879
mango_instrument	Q/Q̃	4	0.936	0.933	0.905	0.864
mango_instrument	Q/Q̃	5	0.919	0.916	0.896	0.840
mango_instrument	Q/Q̃	6	0.919	0.915	0.910	0.839
mango_instrument	Q/Q̃	7	0.911	0.906	0.921	0.831
mango_instrument	Q/Q̃	8	0.871	0.866	0.919	0.774
mango_instrument	Q/Q̃	9	0.859	0.853	0.909	0.761
mango_instrument	Q/Q̃	10	0.827	0.819	0.923	0.691
mango_instrument	T^2	1	0.922	0.918	0.853	0.854
mango_instrument	T^2	2	0.931	0.928	0.887	0.868
mango_instrument	T^2	3	0.938	0.933	0.900	0.875
mango_instrument	T^2	4	0.940	0.936	0.916	0.872
mango_instrument	T^2	5	0.945	0.942	0.930	0.884
mango_instrument	T^2	6	0.944	0.941	0.933	0.883
mango_instrument	T^2	7	0.933	0.929	0.930	0.857
mango_instrument	T^2	8	0.929	0.926	0.934	0.841
mango_instrument	T^2	9	0.916	0.912	0.940	0.805
mango_instrument	T^2	10	0.712	0.700	0.883	0.535
mango_season_forward	Q/Q̃	1	0.971	0.963	0.918	0.913
mango_season_forward	Q/Q̃	2	0.970	0.963	0.932	0.903
mango_season_forward	Q/Q̃	3	0.952	0.943	0.919	0.882
mango_season_forward	Q/Q̃	4	0.946	0.935	0.913	0.855
mango_season_forward	Q/Q̃	5	0.949	0.942	0.934	0.859
mango_season_forward	Q/Q̃	6	0.938	0.928	0.932	0.834
mango_season_forward	Q/Q̃	7	0.920	0.910	0.926	0.821
mango_season_forward	Q/Q̃	8	0.921	0.914	0.934	0.812
mango_season_forward	Q/Q̃	9	0.923	0.917	0.955	0.826
mango_season_forward	Q/Q̃	10	0.626	0.614	0.837	0.520
mango_season_forward	T^2	1	0.951	0.940	0.910	0.877
mango_season_forward	T^2	2	0.948	0.938	0.925	0.882
mango_season_forward	T^2	3	0.939	0.929	0.922	0.865
mango_season_forward	T^2	4	0.925	0.915	0.917	0.841

task	by	decile	ASTM-G	G-W-A	GC-D	SCP
mango_season_forward	T²	5	0.910	0.899	0.915	0.816
mango_season_forward	T²	6	0.905	0.897	0.923	0.823
mango_season_forward	T²	7	0.893	0.885	0.917	0.791
mango_season_forward	T²	8	0.897	0.888	0.917	0.765
mango_season_forward	T²	9	0.895	0.888	0.927	0.786
mango_season_forward	T²	10	0.854	0.852	0.926	0.777
mango_season_loso	Q/Q̃	1	0.985	0.982	0.946	0.947
mango_season_loso	Q/Q̃	2	0.964	0.960	0.925	0.904
mango_season_loso	Q/Q̃	3	0.952	0.947	0.921	0.891
mango_season_loso	Q/Q̃	4	0.945	0.939	0.919	0.880
mango_season_loso	Q/Q̃	5	0.939	0.934	0.920	0.861
mango_season_loso	Q/Q̃	6	0.927	0.921	0.917	0.844
mango_season_loso	Q/Q̃	7	0.930	0.924	0.929	0.852
mango_season_loso	Q/Q̃	8	0.941	0.937	0.948	0.859
mango_season_loso	Q/Q̃	9	0.939	0.933	0.959	0.857
mango_season_loso	Q/Q̃	10	0.788	0.780	0.888	0.694
mango_season_loso	T²	1	0.966	0.963	0.918	0.924
mango_season_loso	T²	2	0.964	0.960	0.930	0.913
mango_season_loso	T²	3	0.958	0.953	0.929	0.903
mango_season_loso	T²	4	0.951	0.946	0.928	0.884
mango_season_loso	T²	5	0.944	0.940	0.926	0.875
mango_season_loso	T²	6	0.945	0.940	0.933	0.872
mango_season_loso	T²	7	0.946	0.941	0.942	0.870
mango_season_loso	T²	8	0.942	0.936	0.946	0.862
mango_season_loso	T²	9	0.916	0.910	0.940	0.820
mango_season_loso	T²	10	0.777	0.769	0.881	0.664
ossl_kssl_to_lucas	Q/Q̃	1	0.102	0.097	0.470	0.065
ossl_kssl_to_lucas	Q/Q̃	2	0.074	0.069	0.474	0.045
ossl_kssl_to_lucas	Q/Q̃	3	0.062	0.058	0.483	0.037
ossl_kssl_to_lucas	Q/Q̃	4	0.054	0.052	0.487	0.035
ossl_kssl_to_lucas	Q/Q̃	5	0.048	0.044	0.517	0.025
ossl_kssl_to_lucas	Q/Q̃	6	0.051	0.045	0.554	0.029
ossl_kssl_to_lucas	Q/Q̃	7	0.051	0.047	0.610	0.030
ossl_kssl_to_lucas	Q/Q̃	8	0.053	0.047	0.656	0.027
ossl_kssl_to_lucas	Q/Q̃	9	0.063	0.059	0.750	0.039
ossl_kssl_to_lucas	Q/Q̃	10	0.098	0.090	0.772	0.059
ossl_kssl_to_lucas	T²	1	0.169	0.163	0.533	0.114
ossl_kssl_to_lucas	T²	2	0.103	0.097	0.544	0.064
ossl_kssl_to_lucas	T²	3	0.084	0.079	0.541	0.050
ossl_kssl_to_lucas	T²	4	0.068	0.062	0.518	0.035
ossl_kssl_to_lucas	T²	5	0.053	0.050	0.543	0.030
ossl_kssl_to_lucas	T²	6	0.050	0.046	0.558	0.028
ossl_kssl_to_lucas	T²	7	0.040	0.036	0.592	0.023
ossl_kssl_to_lucas	T²	8	0.033	0.029	0.628	0.016
ossl_kssl_to_lucas	T²	9	0.032	0.029	0.659	0.019

task	by	decile	ASTM-G	G-W-A	GC-D	SCP
ossl_kssl_to_lucas	T²	10	0.023	0.019	0.654	0.012
ossl_lucas_block	Q/Q̃	1	0.958	0.953	0.895	0.942
ossl_lucas_block	Q/Q̃	2	0.938	0.932	0.895	0.920
ossl_lucas_block	Q/Q̃	3	0.932	0.927	0.891	0.912
ossl_lucas_block	Q/Q̃	4	0.923	0.918	0.894	0.906
ossl_lucas_block	Q/Q̃	5	0.922	0.916	0.903	0.904
ossl_lucas_block	Q/Q̃	6	0.907	0.900	0.894	0.885
ossl_lucas_block	Q/Q̃	7	0.901	0.894	0.895	0.878
ossl_lucas_block	Q/Q̃	8	0.894	0.887	0.898	0.870
ossl_lucas_block	Q/Q̃	9	0.884	0.877	0.907	0.854
ossl_lucas_block	Q/Q̃	10	0.827	0.819	0.907	0.788
ossl_lucas_block	T²	1	0.964	0.961	0.908	0.947
ossl_lucas_block	T²	2	0.938	0.932	0.887	0.919
ossl_lucas_block	T²	3	0.921	0.914	0.878	0.897
ossl_lucas_block	T²	4	0.925	0.918	0.898	0.905
ossl_lucas_block	T²	5	0.916	0.911	0.898	0.898
ossl_lucas_block	T²	6	0.906	0.899	0.892	0.888
ossl_lucas_block	T²	7	0.906	0.899	0.900	0.885
ossl_lucas_block	T²	8	0.907	0.902	0.911	0.886
ossl_lucas_block	T²	9	0.894	0.887	0.914	0.867
ossl_lucas_block	T²	10	0.810	0.801	0.892	0.767

S22. Bias correction with labelled reference samples of the new group

Table S22. k labelled units (fruit) of the new group: no correction (none); interval centre shifted by their mean residual (shifted); GC-D-BC (centre shifted and scale model and q̂ recalibrated on calibration residuals corrected in the same way). Evaluated on the remaining units of each group; 20 random draws; group-averaged coverage, 10th percentile and minimum of the group coverages, interval score and width (seed 0).

task	variant	k	coverage	coverage_q10	worst_group	interval_score	width	n_groups
mango_instrument	GC-D (no correction)	1	0.879	0.791	0.356	6.211	4.542	24
mango_instrument	GC-D (no correction)	2	0.879	0.791	0.357	6.211	4.542	24
mango_instrument	GC-D (no correction)	3	0.879	0.791	0.356	6.212	4.542	24
mango_instrument	GC-D (no correction)	5	0.879	0.792	0.356	6.209	4.542	24
mango_instrument	GC-D (no correction)	10	0.879	0.791	0.356	6.211	4.542	24
mango_instrument	GC-D, centre shifted	1	0.847	0.744	0.671	7.505	4.542	24
mango_instrument	GC-D, centre shifted	2	0.893	0.783	0.724	6.129	4.542	24
mango_instrument	GC-D,	3	0.908	0.814	0.755	5.804	4.542	24

task	variant	k	coverage	coverage_q10	worst_group	interval_score	width	n_groups
	centre shifted							
mango_instrument	GC-D, centre shifted	5	0.917	0.841	0.745	5.650	4.542	24
mango_instrument	GC-D, centre shifted	10	0.936	0.880	0.772	5.353	4.542	24
mango_instrument	GC-D-BC (shifted and recalibrated)	1	0.882	0.814	0.681	7.457	5.032	24
mango_instrument	GC-D-BC (shifted and recalibrated)	2	0.886	0.797	0.724	6.195	4.409	24
mango_instrument	GC-D-BC (shifted and recalibrated)	3	0.882	0.777	0.729	5.898	4.179	24
mango_instrument	GC-D-BC (shifted and recalibrated)	5	0.877	0.778	0.689	5.672	3.938	24
mango_instrument	GC-D-BC (shifted and recalibrated)	10	0.883	0.796	0.690	5.273	3.740	24
mango_population	GC-D (no correction)	1	0.899	0.732	0.012	4.916	3.681	199
mango_population	GC-D (no correction)	2	0.899	0.731	0.012	4.917	3.681	199
mango_population	GC-D (no correction)	3	0.899	0.733	0.014	4.917	3.681	199
mango_population	GC-D (no correction)	5	0.899	0.736	0.013	4.910	3.681	199
mango_population	GC-D (no correction)	10	0.898	0.733	0.009	4.924	3.683	198
mango_population	GC-D, centre shifted	1	0.920	0.824	0.657	4.671	3.681	199
mango_population	GC-D, centre shifted	2	0.946	0.875	0.682	4.248	3.681	199
mango_population	GC-D, centre shifted	3	0.955	0.886	0.684	4.142	3.681	199
mango_population	GC-D, centre shifted	5	0.961	0.908	0.719	4.060	3.681	199
mango_population	GC-D, centre shifted	10	0.966	0.913	0.726	4.019	3.683	198
mango_population	GC-D-BC (shifted and recalibrated)	1	0.900	0.790	0.618	4.646	3.417	199

task	variant	k	coverage	coverage_q10	worst_group	interval_score	width	n_groups
mango_population	GC-D-BC (shifted and recalibrated)	2	0.900	0.799	0.585	4.093	3.058	199
mango_population	GC-D-BC (shifted and recalibrated)	3	0.901	0.796	0.552	3.911	2.924	199
mango_population	GC-D-BC (shifted and recalibrated)	5	0.899	0.798	0.565	3.751	2.802	199
mango_population	GC-D-BC (shifted and recalibrated)	10	0.900	0.796	0.544	3.649	2.725	198
mango_season_loso	GC-D (no correction)	1	0.895	0.819	0.798	5.899	4.519	7
mango_season_loso	GC-D (no correction)	2	0.895	0.819	0.798	5.899	4.519	7
mango_season_loso	GC-D (no correction)	3	0.895	0.819	0.798	5.901	4.519	7
mango_season_loso	GC-D (no correction)	5	0.895	0.819	0.798	5.899	4.519	7
mango_season_loso	GC-D (no correction)	10	0.895	0.819	0.798	5.901	4.519	7
mango_season_loso	GC-D, centre shifted	1	0.795	0.688	0.639	8.397	4.519	7
mango_season_loso	GC-D, centre shifted	2	0.838	0.753	0.702	7.128	4.519	7
mango_season_loso	GC-D, centre shifted	3	0.866	0.785	0.727	6.456	4.519	7
mango_season_loso	GC-D, centre shifted	5	0.873	0.806	0.747	6.355	4.519	7
mango_season_loso	GC-D, centre shifted	10	0.894	0.829	0.800	5.899	4.519	7
mango_season_loso	GC-D-BC (shifted and recalibrated)	1	0.872	0.795	0.759	8.143	5.757	7
mango_season_loso	GC-D-BC (shifted and recalibrated)	2	0.885	0.808	0.778	6.990	5.191	7
mango_season_loso	GC-D-BC (shifted and recalibrated)	3	0.898	0.819	0.776	6.509	5.026	7
mango_season_loso	GC-D-BC (shifted and recalibrated)	5	0.885	0.819	0.765	6.373	4.733	7
mango_season_loso	GC-D-BC (shifted and	10	0.893	0.823	0.807	5.945	4.530	7

task	variant	k	coverage	coverage_q10	worst_group	interval_score	width	n_groups
	recalibrated)							

S23. Q-based rejection rule under test-time perturbations

Table S23. Share of new-season test spectra rejected by the rules Q/Q̃ > 95 % or 99 % quantile of the out-of-group calibration values (and, in rej_Q99_T2, also T^2 > its 99 % quantile), coverage of GC-D on all and on the retained spectra, coverage of SCP and RMSEP (seed 0). For the unperturbed spectra ('none') the rejected share is the false-rejection rate.

perturbation	level	rule	rejected	coverage_all	coverage_retained	scp_coverage_all	rmse_retained	rmse_all
gain	0.050	rej_Q95	0.084	0.928	0.934	0.851	1.216	1.327
gain	0.050	rej_Q99	0.032	0.928	0.932	0.851	1.257	1.327
gain	0.050	rej_Q99_T2	0.040	0.928	0.933	0.851	1.235	1.327
noise	0.003	rej_Q95	0.238	0.894	0.897	0.757	1.503	1.607
noise	0.003	rej_Q99	0.052	0.894	0.898	0.757	1.550	1.607
noise	0.003	rej_Q99_T2	0.059	0.894	0.899	0.757	1.532	1.607
noise	0.010	rej_Q95	0.965	0.704	0.765	0.412	2.639	3.431
noise	0.010	rej_Q99	0.706	0.704	0.741	0.412	3.036	3.431
noise	0.010	rej_Q99_T2	0.715	0.704	0.749	0.412	2.932	3.431
noise	0.030	rej_Q95	1.000	0.448	–	0.153	–	9.625
noise	0.030	rej_Q99	1.000	0.448	0.667	0.153	4.982	9.625
noise	0.030	rej_Q99_T2	1.000	0.448	0.500	0.153	5.156	9.625
none	0.000	rej_Q95	0.069	0.927	0.933	0.859	1.195	1.295
none	0.000	rej_Q99	0.027	0.927	0.931	0.859	1.240	1.295
none	0.000	rej_Q99_T2	0.035	0.927	0.932	0.859	1.215	1.295
offset_slope	2.000	rej_Q95	0.069	0.927	0.933	0.859	1.195	1.295
offset_slope	2.000	rej_Q99	0.027	0.927	0.931	0.859	1.240	1.295
offset_slope	2.000	rej_Q99_T2	0.035	0.927	0.932	0.859	1.215	1.295
wl_shift	0.500	rej_Q95	0.086	0.929	0.932	0.853	1.224	1.305
wl_shift	0.500	rej_Q99	0.028	0.929	0.931	0.853	1.264	1.305
wl_shift	0.500	rej_Q99_T2	0.036	0.929	0.933	0.853	1.233	1.305
wl_shift	1.000	rej_Q95	0.146	0.940	0.940	0.841	1.276	1.337
wl_shift	1.000	rej_Q99	0.041	0.940	0.941	0.841	1.302	1.337
wl_shift	1.000	rej_Q99_T2	0.050	0.940	0.943	0.841	1.274	1.337
wl_shift	2.000	rej_Q95	0.500	0.959	0.953	0.799	1.459	1.460
wl_shift	2.000	rej_Q99	0.177	0.959	0.958	0.799	1.452	1.460
wl_shift	2.000	rej_Q99_T2	0.184	0.959	0.959	0.799	1.435	1.460

S24. Flags for uninformative intervals

Table S24. Share of GC-D test spectra flagged because the interval is at least as wide as the central 90 % range of the training reference values, because Q/Q̃ exceeds the 99 % calibration quantile, or either; coverage and RMSEP of the flagged and retained spectra (seed 0).

task	rule	share_flagged	coverage_all	coverage_retained	coverage_flagged	rmse_retained	rmse_flagged
mango_season_loso	width ≥ reference range	0.009	0.927	0.929	0.747	1.260	3.419
mango_season_loso	Q/Q̃ > 99 % limit	0.027	0.927	0.931	0.804	1.240	2.595
mango_season_loso	either	0.034	0.927	0.931	0.813	1.218	2.680
mango_season_forward	width ≥ reference range	0.010	0.920	0.922	0.692	1.306	4.590
mango_season_forward	Q/Q̃ > 99 % limit	0.050	0.920	0.928	0.774	1.267	2.703
mango_season_forward	either	0.056	0.920	0.929	0.771	1.210	3.018
mango_instrument	width ≥ reference range	0.011	0.911	0.910	0.991	1.363	2.158
mango_instrument	Q/Q̃ > 99 % limit	0.012	0.911	0.910	0.974	1.367	1.846
mango_instrument	either	0.020	0.911	0.909	0.982	1.359	1.963
mango_population	width ≥ reference range	0.000	0.896	0.896	0.000	1.152	8.872
mango_population	Q/Q̃ > 99 % limit	0.009	0.896	0.896	0.909	1.146	1.746
mango_population	either	0.009	0.896	0.896	0.909	1.146	1.746
ossl_lucas_block	width ≥ reference range	0.048	0.898	0.901	0.842	2.293	25.615
ossl_lucas_block	Q/Q̃ > 99 % limit	0.010	0.898	0.898	0.894	5.539	25.391
ossl_lucas_block	either	0.056	0.898	0.900	0.854	2.298	23.810
corn_moisture	width ≥ reference range	0.118	0.522	0.484	0.803	0.613	0.508
corn_moisture	Q/Q̃ > 99 % limit	0.290	0.522	0.711	0.057	0.401	0.924
corn_moisture	either	0.385	0.522	0.672	0.281	0.411	0.818
corn_oil	width ≥ reference range	0.168	0.770	0.752	0.861	0.144	0.248
corn_oil	Q/Q̃ > 99 % limit	0.308	0.770	0.735	0.849	0.177	0.140
corn_oil	either	0.422	0.770	0.715	0.846	0.151	0.185
corn_protein	width ≥ reference	0.210	0.710	0.635	0.992	0.402	0.346

task	rule	share_flagged	coverage_all	coverage_retained	coverage_flagged	rmse_retained	rmse_flagged
	range						
corn_protein	Q/Q̃ > 99 % limit	0.305	0.710	0.739	0.645	0.362	0.451
corn_protein	either	0.483	0.710	0.652	0.772	0.377	0.405
corn_starch	width ≥ reference range	0.452	0.692	0.502	0.923	1.059	0.875
corn_starch	Q/Q̃ > 99 % limit	0.273	0.692	0.842	0.293	0.840	1.281
corn_starch	either	0.650	0.692	0.724	0.674	0.916	1.013
ossl_lucas_campaign	width ≥ reference range	0.044	0.895	0.898	0.840	2.638	20.281
ossl_lucas_campaign	Q/Q̃ > 99 % limit	0.013	0.895	0.896	0.828	4.700	15.193
ossl_lucas_campaign	either	0.055	0.895	0.898	0.838	2.644	18.221
ossl_kssl_to_lucas	width ≥ reference range	0.484	0.577	0.681	0.467	6.940	45.731
ossl_kssl_to_lucas	Q/Q̃ > 99 % limit	0.028	0.577	0.573	0.731	30.020	75.651
ossl_kssl_to_lucas	either	0.498	0.577	0.674	0.480	6.991	45.116
tablets	width ≥ reference range	0.000	0.899	0.899	–	4.902	–
tablets	Q/Q̃ > 99 % limit	0.501	0.899	0.932	0.866	4.308	5.430
tablets	either	0.501	0.899	0.932	0.866	4.308	5.430

S25. Specification-limit decisions over a grid of limits

Table S25a. Decisions 'y ≥ L' accepted when the lower interval limit ≥ L (mango, new season; soil, new region; seed 0). point_rule_*: acceptance when ŷ ≥ L.

task	method	L	share_compliant	false_acceptance	yield_compliant	point_rule_false_acceptance	point_rule_yield
mango_season_loso	ASTM-G	12.000	0.957	0.005	0.770	0.033	0.990
mango_season_loso	ASTM-G	12.500	0.926	0.006	0.705	0.052	0.979
mango_season_loso	ASTM-G	13.000	0.883	0.008	0.638	0.071	0.965
mango_season_loso	ASTM-G	13.500	0.822	0.011	0.580	0.096	0.949
mango_season_loso	ASTM-G	14.000	0.756	0.013	0.524	0.116	0.929
mango_season_loso	ASTM-G	14.500	0.683	0.013	0.475	0.132	0.911
mango_season_loso	ASTM-G	15.000	0.610	0.015	0.428	0.135	0.897
mango_season_loso	ASTM-G	15.500	0.545	0.020	0.372	0.131	0.866
mango_season_loso	ASTM-G	16.000	0.470	0.030	0.322	0.145	0.831
mango_season_loso	ASTM-G	16.500	0.399	0.037	0.272	0.160	0.796
mango_season_loso	ASTM-G	17.000	0.329	0.043	0.229	0.178	0.770

task	method	L	share_compliant	false_acceptance	yield_compliant	point_rule_false_acceptance	point_rule_yield
mango_season_loso	ASTM-G	17.500	0.264	0.051	0.192	0.204	0.741
mango_season_loso	ASTM-G	18.000	0.208	0.071	0.156	0.242	0.690
mango_season_loso	ASTM-G	18.500	0.150	0.102	0.135	0.299	0.658
mango_season_loso	ASTM-G	19.000	0.106	0.098	0.113	0.355	0.605
mango_season_loso	ASTM-R	12.000	0.957	0.007	0.840	0.033	0.990
mango_season_loso	ASTM-R	12.500	0.926	0.010	0.783	0.052	0.979
mango_season_loso	ASTM-R	13.000	0.883	0.013	0.726	0.071	0.965
mango_season_loso	ASTM-R	13.500	0.822	0.018	0.670	0.096	0.949
mango_season_loso	ASTM-R	14.000	0.756	0.022	0.616	0.116	0.929
mango_season_loso	ASTM-R	14.500	0.683	0.024	0.565	0.132	0.911
mango_season_loso	ASTM-R	15.000	0.610	0.022	0.520	0.135	0.897
mango_season_loso	ASTM-R	15.500	0.545	0.025	0.467	0.131	0.866
mango_season_loso	ASTM-R	16.000	0.470	0.039	0.416	0.145	0.831
mango_season_loso	ASTM-R	16.500	0.399	0.047	0.366	0.160	0.796
mango_season_loso	ASTM-R	17.000	0.329	0.056	0.318	0.178	0.770
mango_season_loso	ASTM-R	17.500	0.264	0.072	0.271	0.204	0.741
mango_season_loso	ASTM-R	18.000	0.208	0.097	0.227	0.242	0.690
mango_season_loso	ASTM-R	18.500	0.150	0.128	0.201	0.299	0.658
mango_season_loso	ASTM-R	19.000	0.106	0.143	0.178	0.355	0.605
mango_season_loso	CQR	12.000	0.957	0.013	0.901	0.033	0.990
mango_season_loso	CQR	12.500	0.926	0.016	0.845	0.052	0.979
mango_season_loso	CQR	13.000	0.883	0.019	0.787	0.071	0.965
mango_season_loso	CQR	13.500	0.822	0.024	0.728	0.096	0.949
mango_season_loso	CQR	14.000	0.756	0.027	0.665	0.116	0.929
mango_season_loso	CQR	14.500	0.683	0.024	0.601	0.132	0.911
mango_season_loso	CQR	15.000	0.610	0.013	0.535	0.135	0.897
mango_season_loso	CQR	15.500	0.545	0.009	0.448	0.131	0.866
mango_season_loso	CQR	16.000	0.470	0.017	0.351	0.145	0.831
mango_season_loso	CQR	16.500	0.399	0.015	0.243	0.160	0.796
mango_season_loso	CQR	17.000	0.329	0.007	0.146	0.178	0.770
mango_season_loso	CQR	17.500	0.264	0.006	0.069	0.204	0.741
mango_season_loso	CQR	18.000	0.208	0.000	0.023	0.242	0.690
mango_season_loso	CQR	18.500	0.150	0.000	0.008	0.299	0.658
mango_season_loso	CQR	19.000	0.106	0.000	0.001	0.355	0.605
mango_season_loso	GC-D	12.000	0.957	0.006	0.783	0.033	0.990
mango_season_loso	GC-D	12.500	0.926	0.007	0.722	0.052	0.979
mango_season_loso	GC-D	13.000	0.883	0.008	0.662	0.071	0.965
mango_season_loso	GC-D	13.500	0.822	0.011	0.608	0.096	0.949
mango_season_loso	GC-D	14.000	0.756	0.013	0.551	0.116	0.929
mango_season_loso	GC-D	14.500	0.683	0.012	0.502	0.132	0.911
mango_season_loso	GC-D	15.000	0.610	0.009	0.453	0.135	0.897
mango_season_loso	GC-D	15.500	0.545	0.012	0.397	0.131	0.866
mango_season_loso	GC-D	16.000	0.470	0.021	0.342	0.145	0.831

task	method	L	share_compliant	false_acceptance	yield_compliant	point_rule_false_acceptance	point_rule_yield
mango_season_loso	GC-D	16.500	0.399	0.023	0.289	0.160	0.796
mango_season_loso	GC-D	17.000	0.329	0.024	0.240	0.178	0.770
mango_season_loso	GC-D	17.500	0.264	0.033	0.196	0.204	0.741
mango_season_loso	GC-D	18.000	0.208	0.049	0.160	0.242	0.690
mango_season_loso	GC-D	18.500	0.150	0.070	0.134	0.299	0.658
mango_season_loso	GC-D	19.000	0.106	0.061	0.114	0.355	0.605
mango_season_loso	ORACLE	12.000	0.957	0.006	0.817	0.033	0.990
mango_season_loso	ORACLE	12.500	0.926	0.008	0.762	0.052	0.979
mango_season_loso	ORACLE	13.000	0.883	0.010	0.704	0.071	0.965
mango_season_loso	ORACLE	13.500	0.822	0.014	0.649	0.096	0.949
mango_season_loso	ORACLE	14.000	0.756	0.015	0.591	0.116	0.929
mango_season_loso	ORACLE	14.500	0.683	0.015	0.538	0.132	0.911
mango_season_loso	ORACLE	15.000	0.610	0.016	0.485	0.135	0.897
mango_season_loso	ORACLE	15.500	0.545	0.018	0.427	0.131	0.866
mango_season_loso	ORACLE	16.000	0.470	0.026	0.375	0.145	0.831
mango_season_loso	ORACLE	16.500	0.399	0.034	0.319	0.160	0.796
mango_season_loso	ORACLE	17.000	0.329	0.035	0.272	0.178	0.770
mango_season_loso	ORACLE	17.500	0.264	0.045	0.227	0.204	0.741
mango_season_loso	ORACLE	18.000	0.208	0.074	0.180	0.242	0.690
mango_season_loso	ORACLE	18.500	0.150	0.109	0.148	0.299	0.658
mango_season_loso	ORACLE	19.000	0.106	0.124	0.123	0.355	0.605
mango_season_loso	SCP	12.000	0.957	0.007	0.850	0.033	0.990
mango_season_loso	SCP	12.500	0.926	0.010	0.795	0.052	0.979
mango_season_loso	SCP	13.000	0.883	0.013	0.739	0.071	0.965
mango_season_loso	SCP	13.500	0.822	0.019	0.685	0.096	0.949
mango_season_loso	SCP	14.000	0.756	0.023	0.631	0.116	0.929
mango_season_loso	SCP	14.500	0.683	0.024	0.582	0.132	0.911
mango_season_loso	SCP	15.000	0.610	0.024	0.536	0.135	0.897
mango_season_loso	SCP	15.500	0.545	0.027	0.484	0.131	0.866
mango_season_loso	SCP	16.000	0.470	0.041	0.434	0.145	0.831
mango_season_loso	SCP	16.500	0.399	0.050	0.385	0.160	0.796
mango_season_loso	SCP	17.000	0.329	0.057	0.339	0.178	0.770
mango_season_loso	SCP	17.500	0.264	0.076	0.287	0.204	0.741
mango_season_loso	SCP	18.000	0.208	0.099	0.241	0.242	0.690
mango_season_loso	SCP	18.500	0.150	0.131	0.216	0.299	0.658
mango_season_loso	SCP	19.000	0.106	0.141	0.190	0.355	0.605
ossl_lucas_block	ASTM-G	1.000	0.852	0.015	0.506	0.104	0.958
ossl_lucas_block	ASTM-G	1.500	0.663	0.020	0.420	0.178	0.917
ossl_lucas_block	ASTM-G	2.000	0.516	0.026	0.372	0.214	0.872
ossl_lucas_block	ASTM-G	3.000	0.342	0.024	0.324	0.208	0.785
ossl_lucas_block	ASTM-G	5.000	0.186	0.019	0.323	0.158	0.678
ossl_lucas_block	ASTM-G	8.000	0.107	0.008	0.399	0.092	0.631
ossl_lucas_block	ASTM-R	1.000	0.852	0.016	0.533	0.104	0.958

task	method	L	share_compliant	false_acceptance	yield_compliant	point_rule_false_acceptance	point_rule_yield
ossl_lucas_block	ASTM-R	1.500	0.663	0.022	0.443	0.178	0.917
ossl_lucas_block	ASTM-R	2.000	0.516	0.029	0.395	0.214	0.872
ossl_lucas_block	ASTM-R	3.000	0.342	0.027	0.340	0.208	0.785
ossl_lucas_block	ASTM-R	5.000	0.186	0.022	0.333	0.158	0.678
ossl_lucas_block	ASTM-R	8.000	0.107	0.007	0.407	0.092	0.631
ossl_lucas_block	CQR	1.000	0.852	0.015	0.616	0.104	0.958
ossl_lucas_block	CQR	1.500	0.663	0.012	0.388	0.178	0.917
ossl_lucas_block	CQR	2.000	0.516	0.010	0.287	0.214	0.872
ossl_lucas_block	CQR	3.000	0.342	0.008	0.214	0.208	0.785
ossl_lucas_block	CQR	5.000	0.186	0.006	0.248	0.158	0.678
ossl_lucas_block	CQR	8.000	0.107	0.003	0.316	0.092	0.631
ossl_lucas_block	GC-D	1.000	0.852	0.016	0.543	0.104	0.958
ossl_lucas_block	GC-D	1.500	0.663	0.023	0.449	0.178	0.917
ossl_lucas_block	GC-D	2.000	0.516	0.030	0.395	0.214	0.872
ossl_lucas_block	GC-D	3.000	0.342	0.025	0.331	0.208	0.785
ossl_lucas_block	GC-D	5.000	0.186	0.016	0.322	0.158	0.678
ossl_lucas_block	GC-D	8.000	0.107	0.003	0.390	0.092	0.631
ossl_lucas_block	ORACLE	1.000	0.852	0.015	0.545	0.104	0.958
ossl_lucas_block	ORACLE	1.500	0.663	0.021	0.442	0.178	0.917
ossl_lucas_block	ORACLE	2.000	0.516	0.025	0.381	0.214	0.872
ossl_lucas_block	ORACLE	3.000	0.342	0.022	0.320	0.208	0.785
ossl_lucas_block	ORACLE	5.000	0.186	0.020	0.311	0.158	0.678
ossl_lucas_block	ORACLE	8.000	0.107	0.007	0.382	0.092	0.631
ossl_lucas_block	SCP	1.000	0.852	0.017	0.546	0.104	0.958
ossl_lucas_block	SCP	1.500	0.663	0.024	0.453	0.178	0.917
ossl_lucas_block	SCP	2.000	0.516	0.031	0.404	0.214	0.872
ossl_lucas_block	SCP	3.000	0.342	0.029	0.349	0.208	0.785
ossl_lucas_block	SCP	5.000	0.186	0.022	0.339	0.158	0.678
ossl_lucas_block	SCP	8.000	0.107	0.008	0.412	0.092	0.631

Table S25b. Break-even cost ratio c* (cost of a false acceptance relative to a false rejection) above which GC-D has the lower expected cost than the comparator; empty when GC-D has both fewer false acceptances and fewer false rejections (GC-D preferable for every c) or more of both.

task	L	comparator	delta_false_accept	delta_false_reject	break_even_cost_ratio
mango_season_loso	12.000	SCP	0.002	0.064	37.719
mango_season_loso	12.000	ASTM-R	0.001	0.054	40.456
mango_season_loso	12.000	ASTM-G	-0.001	-0.013	–
mango_season_loso	12.500	SCP	0.003	0.067	23.925
mango_season_loso	12.500	ASTM-R	0.002	0.056	23.067
mango_season_loso	12.500	ASTM-G	-0.001	-0.016	–
mango_season_loso	13.000	SCP	0.004	0.068	17.858
mango_season_loso	13.000	ASTM-R	0.004	0.057	16.007

task	L	comparator	delta_false_accept	delta_false_reject	break_even_cost_ratio
mango_season_loso	13.000	ASTM-G	-0.000	-0.021	–
mango_season_loso	13.500	SCP	0.005	0.064	12.000
mango_season_loso	13.500	ASTM-R	0.005	0.051	10.688
mango_season_loso	13.500	ASTM-G	-0.000	-0.023	–
mango_season_loso	14.000	SCP	0.006	0.060	10.098
mango_season_loso	14.000	ASTM-R	0.005	0.049	9.188
mango_season_loso	14.000	ASTM-G	-0.000	-0.021	–
mango_season_loso	14.500	SCP	0.006	0.055	9.532
mango_season_loso	14.500	ASTM-R	0.005	0.043	8.267
mango_season_loso	14.500	ASTM-G	0.000	-0.018	-76.800
mango_season_loso	15.000	SCP	0.005	0.051	9.283
mango_season_loso	15.000	ASTM-R	0.005	0.041	8.818
mango_season_loso	15.000	ASTM-G	0.001	-0.015	-11.256
mango_season_loso	15.500	SCP	0.005	0.047	10.058
mango_season_loso	15.500	ASTM-R	0.004	0.038	9.426
mango_season_loso	15.500	ASTM-G	0.002	-0.014	-8.840
mango_season_loso	16.000	SCP	0.005	0.043	8.199
mango_season_loso	16.000	ASTM-R	0.004	0.035	7.811
mango_season_loso	16.000	ASTM-G	0.001	-0.009	-7.812
mango_season_loso	16.500	SCP	0.005	0.038	7.087
mango_season_loso	16.500	ASTM-R	0.005	0.031	6.756
mango_season_loso	16.500	ASTM-G	0.001	-0.007	-4.484
mango_season_loso	17.000	SCP	0.005	0.033	6.850
mango_season_loso	17.000	ASTM-R	0.004	0.026	6.156
mango_season_loso	17.000	ASTM-G	0.001	-0.003	-2.496
mango_season_loso	17.500	SCP	0.004	0.024	5.408
mango_season_loso	17.500	ASTM-R	0.004	0.020	5.190
mango_season_loso	17.500	ASTM-G	0.001	-0.001	-1.321
mango_season_loso	18.000	SCP	0.004	0.017	4.421
mango_season_loso	18.000	ASTM-R	0.003	0.014	4.125
mango_season_loso	18.000	ASTM-G	0.001	-0.001	-0.985
mango_season_loso	18.500	SCP	0.003	0.012	3.635
mango_season_loso	18.500	ASTM-R	0.003	0.010	3.474
mango_season_loso	18.500	ASTM-G	0.001	0.000	0.061
mango_season_loso	19.000	SCP	0.003	0.008	3.199
mango_season_loso	19.000	ASTM-R	0.002	0.007	2.867
mango_season_loso	19.000	ASTM-G	0.001	-0.000	-0.136
ossl_lucas_block	1.000	SCP	0.000	0.002	6.308
ossl_lucas_block	1.000	ASTM-R	-0.000	-0.009	–
ossl_lucas_block	1.000	ASTM-G	-0.001	-0.032	–
ossl_lucas_block	1.500	SCP	0.000	0.003	8.417
ossl_lucas_block	1.500	ASTM-R	-0.000	-0.004	–
ossl_lucas_block	1.500	ASTM-G	-0.001	-0.020	–
ossl_lucas_block	2.000	SCP	0.000	0.005	9.950

task	L	comparator	delta_false_accept	delta_false_reject	break_even_cost_ratio
ossl_lucas_block	2.000	ASTM-R	-0.000	0.000	–
ossl_lucas_block	2.000	ASTM-G	-0.001	-0.012	–
ossl_lucas_block	3.000	SCP	0.001	0.006	8.552
ossl_lucas_block	3.000	ASTM-R	0.000	0.003	8.500
ossl_lucas_block	3.000	ASTM-G	-0.000	-0.002	–
ossl_lucas_block	5.000	SCP	0.000	0.003	6.579
ossl_lucas_block	5.000	ASTM-R	0.000	0.002	5.125
ossl_lucas_block	5.000	ASTM-G	0.000	0.000	1.143
ossl_lucas_block	8.000	SCP	0.000	0.002	10.333
ossl_lucas_block	8.000	ASTM-R	0.000	0.002	8.875
ossl_lucas_block	8.000	ASTM-G	0.000	0.001	5.000

S26. Synthetic study of single-deployment coverage

Model: $y = x'\beta + b_g + \varepsilon$ with $p = 10$ features, $x \sim N(\mu_g, I)$, group mean $\mu_g \sim N(0, I)$, group offset $b_g \sim N(0, \tau^2)$ and heteroscedastic noise $\varepsilon \sim N(0, 1 + \|x\|^2/p)$. A least-squares model fitted on K calibration groups (50 observations each, or log-normally distributed sizes) is deployed on a new group of 200 observations; 1000 deployments per setting. Because the group means μ_g let the model partly fit the offsets of the calibration groups, random-split residuals under-state the error for a new group, as for real spectra. GC-A: leave-one-group-out absolute residuals with group-balanced weights; GC-A+: with the finite-sample point mass.

Table S26. Coverage attained in one new group (1000 deployments per setting): mean, standard deviation and share of deployments covered more than 0.05 below nominal for random-split calibration (SCP), out-of-group calibration (GC-A) and GC-A with the finite-sample point mass (GC-A+), and the share of finite GC-A+ intervals. Further statistics (10th percentile, between-group share of the variance) in results/revision/simulation_finiteK.csv.

alpha	tau	balance	K	mean GC-A	mean GC-A+	mean SCP	SD GC-A	SD GC-A+	SD SCP	share< −0.05 GC-A	share< −0.05 GC-A+	share< −0.05 SCP	GC-A+ finite
0.050	0.500	balanced	3	0.947	1.000	0.944	0.049	0.000	0.059	0.127	0.000	0.166	0.000
0.050	0.500	balanced	5	0.951	1.000	0.948	0.036	0.000	0.042	0.079	0.000	0.096	0.000
0.050	0.500	balanced	10	0.947	1.000	0.945	0.030	0.000	0.036	0.073	0.000	0.101	0.000
0.050	0.500	balanced	20	0.950	0.997	0.949	0.026	0.005	0.029	0.042	0.000	0.052	1.000
0.050	0.500	balanced	50	0.949	0.968	0.948	0.026	0.020	0.028	0.050	0.004	0.056	1.000
0.050	0.500	unbalanced	3	0.954	1.000	0.955	0.053	0.000	0.055	0.099	0.000	0.137	0.000
0.050	0.500	unbalanced	5	0.948	1.000	0.947	0.040	0.000	0.048	0.094	0.000	0.133	0.000
0.050	0.500	unbalanced	10	0.949	1.000	0.945	0.032	0.000	0.038	0.077	0.000	0.110	0.000
0.050	0.500	unbalanced	20	0.948	0.996	0.945	0.030	0.006	0.034	0.062	0.000	0.086	1.000

alpha	tau	balance	K	mean GC-A	mean GC-A+	mean SCP	SD GC-A	SD GC-A+	SD SCP	share< −0.05 GC-A	share< −0.05 GC-A+	share< −0.05 SCP	GC-A+ finite
0.050	0.500	unbalanced	50	0.950	0.969	0.949	0.025	0.019	0.026	0.034	0.004	0.051	1.000
0.050	1.000	balanced	3	0.942	1.000	0.925	0.068	0.000	0.089	0.152	0.000	0.233	0.000
0.050	1.000	balanced	5	0.938	1.000	0.919	0.069	0.000	0.083	0.166	0.000	0.250	0.000
0.050	1.000	balanced	10	0.947	1.000	0.931	0.054	0.000	0.065	0.128	0.000	0.200	0.000
0.050	1.000	balanced	20	0.947	0.996	0.934	0.049	0.008	0.059	0.112	0.000	0.164	1.000
0.050	1.000	balanced	50	0.952	0.971	0.947	0.043	0.031	0.045	0.086	0.034	0.097	1.000
0.050	1.000	unbalanced	3	0.947	1.000	0.937	0.075	0.000	0.090	0.159	0.000	0.207	0.000
0.050	1.000	unbalanced	5	0.939	1.000	0.923	0.064	0.000	0.076	0.188	0.000	0.242	0.000
0.050	1.000	unbalanced	10	0.946	1.000	0.926	0.053	0.000	0.067	0.138	0.000	0.213	0.000
0.050	1.000	unbalanced	20	0.948	0.996	0.930	0.050	0.009	0.060	0.120	0.002	0.196	1.000
0.050	1.000	unbalanced	50	0.947	0.967	0.937	0.050	0.036	0.055	0.117	0.042	0.158	1.000
0.100	0.500	balanced	3	0.902	1.000	0.883	0.063	0.000	0.080	0.158	0.000	0.257	0.000
0.100	0.500	balanced	5	0.897	1.000	0.884	0.054	0.000	0.065	0.158	0.000	0.259	0.000
0.100	0.500	balanced	10	0.899	0.988	0.892	0.045	0.013	0.050	0.113	0.000	0.177	1.000
0.100	0.500	balanced	20	0.902	0.946	0.899	0.037	0.027	0.040	0.082	0.004	0.105	1.000
0.100	0.500	balanced	50	0.901	0.919	0.899	0.037	0.033	0.039	0.088	0.028	0.093	1.000
0.100	0.500	unbalanced	3	0.915	1.000	0.887	0.072	0.000	0.097	0.133	0.000	0.265	0.000
0.100	0.500	unbalanced	5	0.899	1.000	0.886	0.058	0.000	0.067	0.156	0.000	0.242	0.000
0.100	0.500	unbalanced	10	0.896	0.987	0.887	0.048	0.014	0.055	0.146	0.000	0.202	1.000
0.100	0.500	unbalanced	20	0.897	0.942	0.891	0.043	0.032	0.045	0.108	0.013	0.164	1.000
0.100	0.500	unbalanced	50	0.899	0.918	0.896	0.038	0.035	0.040	0.093	0.039	0.114	1.000
0.100	1.000	balanced	3	0.885	1.000	0.841	0.105	0.000	0.128	0.250	0.000	0.393	0.000
0.100	1.000	balanced	5	0.889	1.000	0.852	0.091	0.000	0.105	0.222	0.000	0.354	0.000
0.100	1.000	balanced	10	0.899	0.987	0.866	0.076	0.022	0.088	0.177	0.004	0.305	1.000

alpha	tau	balance	K	mean GC-A	mean GC-A+	mean SCP	SD GC-A	SD GC-A+	SD SCP	share< −0.05 GC-A	share< −0.05 GC-A+	share< −0.05 SCP	GC-A+ finite
0.100	1.000	balanced	20	0.900	0.944	0.881	0.069	0.048	0.075	0.168	0.052	0.225	1.000
0.100	1.000	balanced	50	0.902	0.919	0.894	0.069	0.062	0.074	0.136	0.089	0.154	1.000
0.100	1.000	unbalanced	3	0.900	1.000	0.851	0.108	0.000	0.135	0.217	0.000	0.360	0.000
0.100	1.000	unbalanced	5	0.892	1.000	0.850	0.089	0.000	0.103	0.223	0.000	0.372	0.000
0.100	1.000	unbalanced	10	0.898	0.986	0.860	0.079	0.024	0.090	0.171	0.003	0.326	1.000
0.100	1.000	unbalanced	20	0.895	0.941	0.868	0.079	0.057	0.089	0.165	0.056	0.277	1.000
0.100	1.000	unbalanced	50	0.900	0.918	0.884	0.071	0.064	0.077	0.161	0.111	0.215	1.000
0.200	0.500	balanced	3	0.814	1.000	0.778	0.086	0.000	0.098	0.198	0.000	0.312	0.000
0.200	0.500	balanced	5	0.802	0.956	0.783	0.070	0.033	0.081	0.193	0.001	0.297	1.000
0.200	0.500	balanced	10	0.800	0.880	0.788	0.060	0.048	0.067	0.178	0.010	0.258	1.000
0.200	0.500	balanced	20	0.799	0.838	0.792	0.053	0.050	0.058	0.171	0.046	0.215	1.000
0.200	0.500	balanced	50	0.798	0.813	0.794	0.051	0.050	0.054	0.150	0.099	0.175	1.000
0.200	0.500	unbalanced	3	0.826	1.000	0.777	0.107	0.000	0.116	0.195	0.000	0.340	0.000
0.200	0.500	unbalanced	5	0.808	0.959	0.781	0.076	0.036	0.088	0.194	0.003	0.303	1.000
0.200	0.500	unbalanced	10	0.800	0.878	0.782	0.062	0.051	0.071	0.183	0.019	0.278	1.000
0.200	0.500	unbalanced	20	0.798	0.838	0.787	0.056	0.052	0.059	0.175	0.055	0.230	1.000
0.200	0.500	unbalanced	50	0.799	0.815	0.792	0.051	0.050	0.054	0.154	0.101	0.194	1.000
0.200	1.000	balanced	3	0.794	1.000	0.718	0.136	0.000	0.147	0.289	0.000	0.488	0.000
0.200	1.000	balanced	5	0.799	0.952	0.739	0.116	0.055	0.126	0.254	0.012	0.440	1.000
0.200	1.000	balanced	10	0.798	0.876	0.753	0.110	0.089	0.117	0.220	0.079	0.369	1.000
0.200	1.000	balanced	20	0.802	0.842	0.776	0.100	0.092	0.105	0.209	0.128	0.269	1.000
0.200	1.000	balanced	50	0.797	0.813	0.787	0.097	0.095	0.099	0.214	0.175	0.247	1.000
0.200	1.000	unbalanced	3	0.822	1.000	0.719	0.147	0.000	0.158	0.225	0.000	0.502	0.000
0.200	1.000	unbalanced	5	0.800	0.951	0.735	0.120	0.058	0.125	0.268	0.015	0.439	1.000

alpha	tau	balance	K	mean GC-A	mean GC-A+	mean SCP	SD GC-A	SD GC-A+	SD SCP	share< −0.05 GC-A	share< −0.05 GC-A+	share< −0.05 SCP	GC-A+ finite
0.200	1.000	unbalanced	10	0.799	0.878	0.750	0.111	0.089	0.117	0.236	0.080	0.401	1.000
0.200	1.000	unbalanced	20	0.803	0.843	0.767	0.095	0.087	0.101	0.215	0.131	0.316	1.000
0.200	1.000	unbalanced	50	0.798	0.814	0.776	0.096	0.093	0.102	0.222	0.182	0.289	1.000

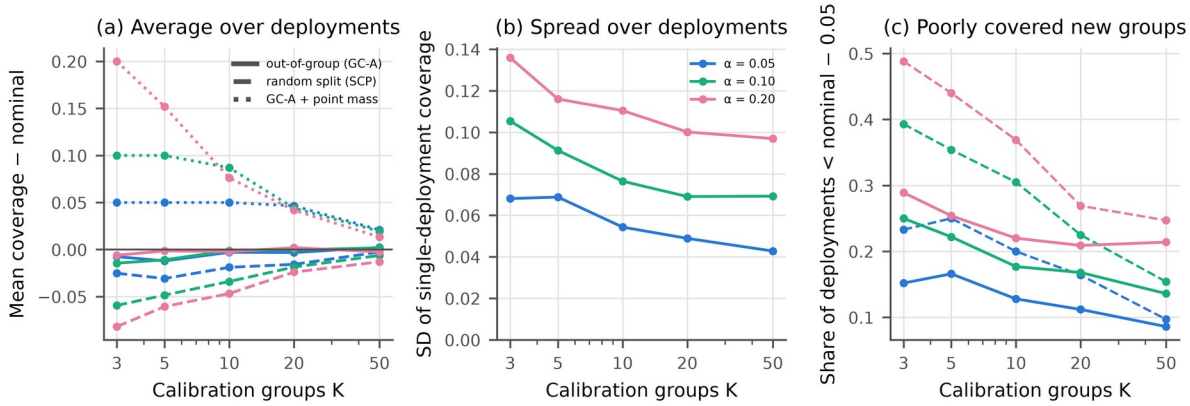


Fig. S3. Synthetic study (τ = 1, balanced groups): (a) mean coverage minus nominal, (b) standard deviation of the coverage in one new group and (c) share of new groups covered more than 0.05 below nominal, as a function of the number of calibration groups K for α = 0.05, 0.10 and 0.20.

S27. Does a new mango population bring additional error?

Table S27a. RMSEP of the final PLS models (GC-D centre, seed 0) for held-out populations, seasons and instruments, compared with random unit-wise 5-fold cross-validation of all mango spectra.

scenario	rmsep	A	ratio_to_random_cv
random unit-wise 5-fold CV (no group shift)	1.105	19.000	1.000
new population	1.153	19.000	1.044
new season	1.295	15.000	1.172
next season	1.376	15.000	1.245
new instrument	1.374	13.000	1.244

Table S27b. Population effects: standard deviation of population-mean residuals when the population is new but its season is in training (population scenario) and when its season is new (LOSO); share of their variance explained by season (R^2); share of the between-population variance of the population-mean PLS scores (A = 15, all spectra) explained by season and by season plus instrument composition.

Quantity	Value
n_populations	199
n_parameters_season_instrument	37
sd_pop_mean_residual_population_scenario	0.805
sd_pop_mean_residual_l	0.889

Quantity	Value
oso_scenario	
r2_season_of_pop_residual_means_loso	0.155
r2_season_of_pop_residual_means_population	0.139
r2_pop_mean_scores_by_season	0.088
r2_pop_mean_scores_by_season_and_instruments	0.401
worst_population	114500
worst_population_bias	-3.233

S28. Group-to-group dispersion of coverage

Table S28. Standard deviation of per-group coverage, the standard deviation expected from sampling alone (binomial, with the number of physical units per group), the share of between-group variance, the 10th percentile of group coverages and the share of groups below 0.85 (seed 0).

task	method	n_groups	mean_coverage	sd_group_coverage	expected_sampling_sd	between_group_share	q10_group_coverage	share_groups_below_085
mango_season_loso	GC-D	7	0.895	0.066	0.010	0.978	0.819	0.286
mango_season_loso	SCP	7	0.823	0.086	0.012	0.982	0.739	0.714
mango_season_forward	GC-D	5	0.877	0.080	0.011	0.982	0.796	0.400
mango_season_forward	SCP	5	0.778	0.086	0.013	0.977	0.706	0.800
mango_instrument	GC-D	24	0.879	0.128	0.011	0.993	0.791	0.292
mango_instrument	SCP	24	0.795	0.167	0.013	0.994	0.630	0.542
mango_population	GC-D	199	0.899	0.150	0.039	0.931	0.732	0.191
mango_population	SCP	199	0.885	0.153	0.042	0.924	0.729	0.251
ossl_lucas_block	GC-D	30	0.898	0.064	0.011	0.971	0.796	0.267
ossl_lucas_block	SCP	30	0.883	0.063	0.012	0.962	0.792	0.267